



TRACK STRUCTURES AND COMPONENTS

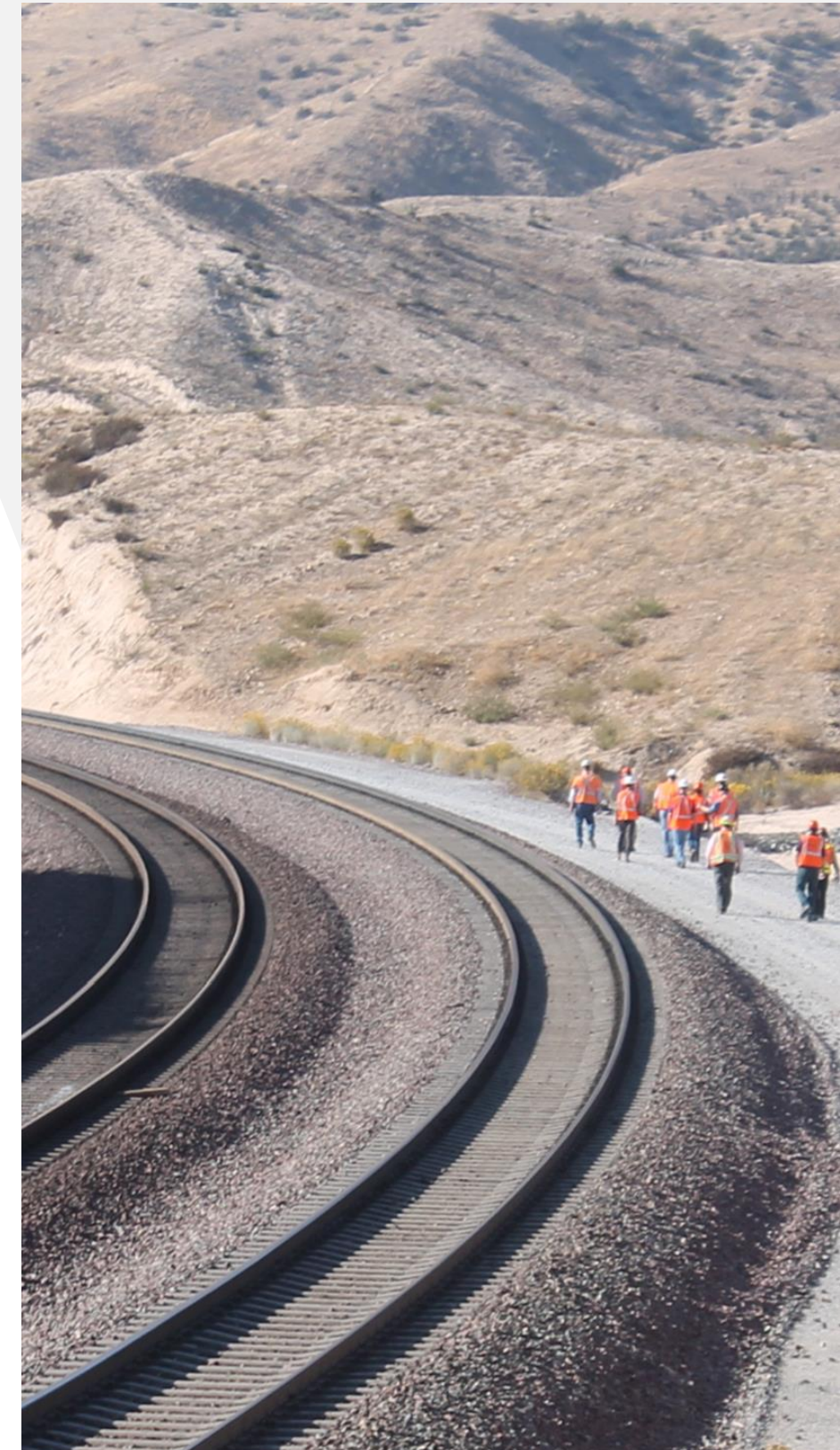
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DALLAS, TX



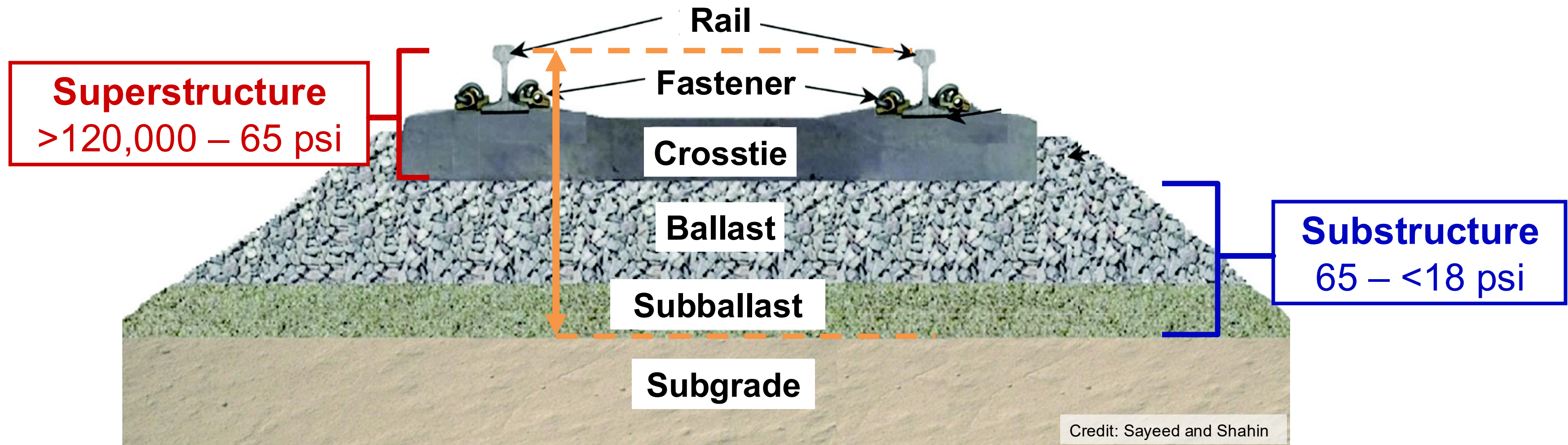
Outline

- Basics of the Track Structure and its Components
- Horizontal Curvature, Superelevation, Spirals, and Track Design
- Track Standards and Regulations
- The Track and Vehicle and as a System
- Track Loading Environment Basics



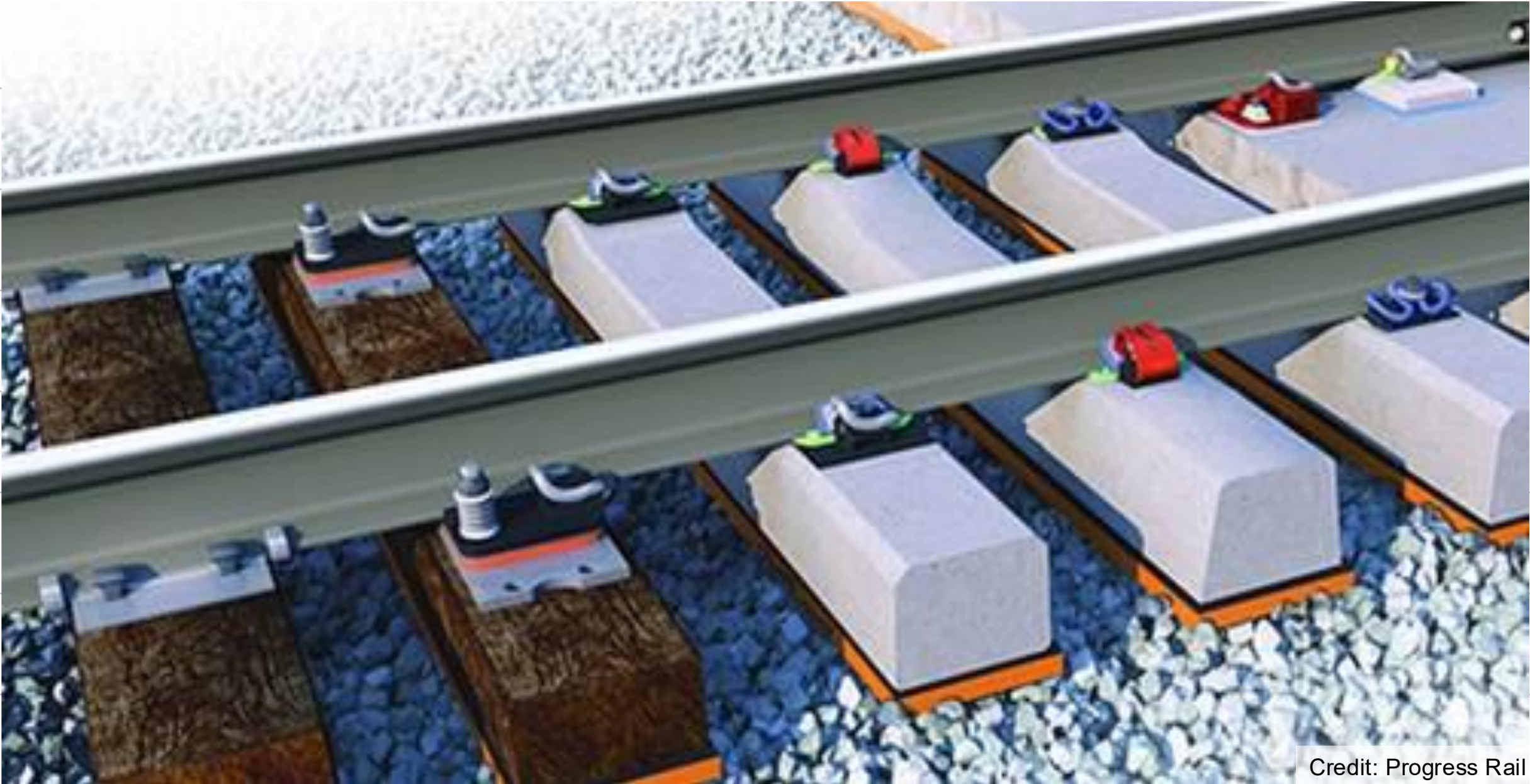
Basics of the Track Structure and its Components

Track Superstructure & Substructure



- **Typical Track System Depth: 29 to 39 inches**
- Material selection and quality are optimized for each step in load transfer
- Component interfaces are where the “load transfer” battle is won and lost

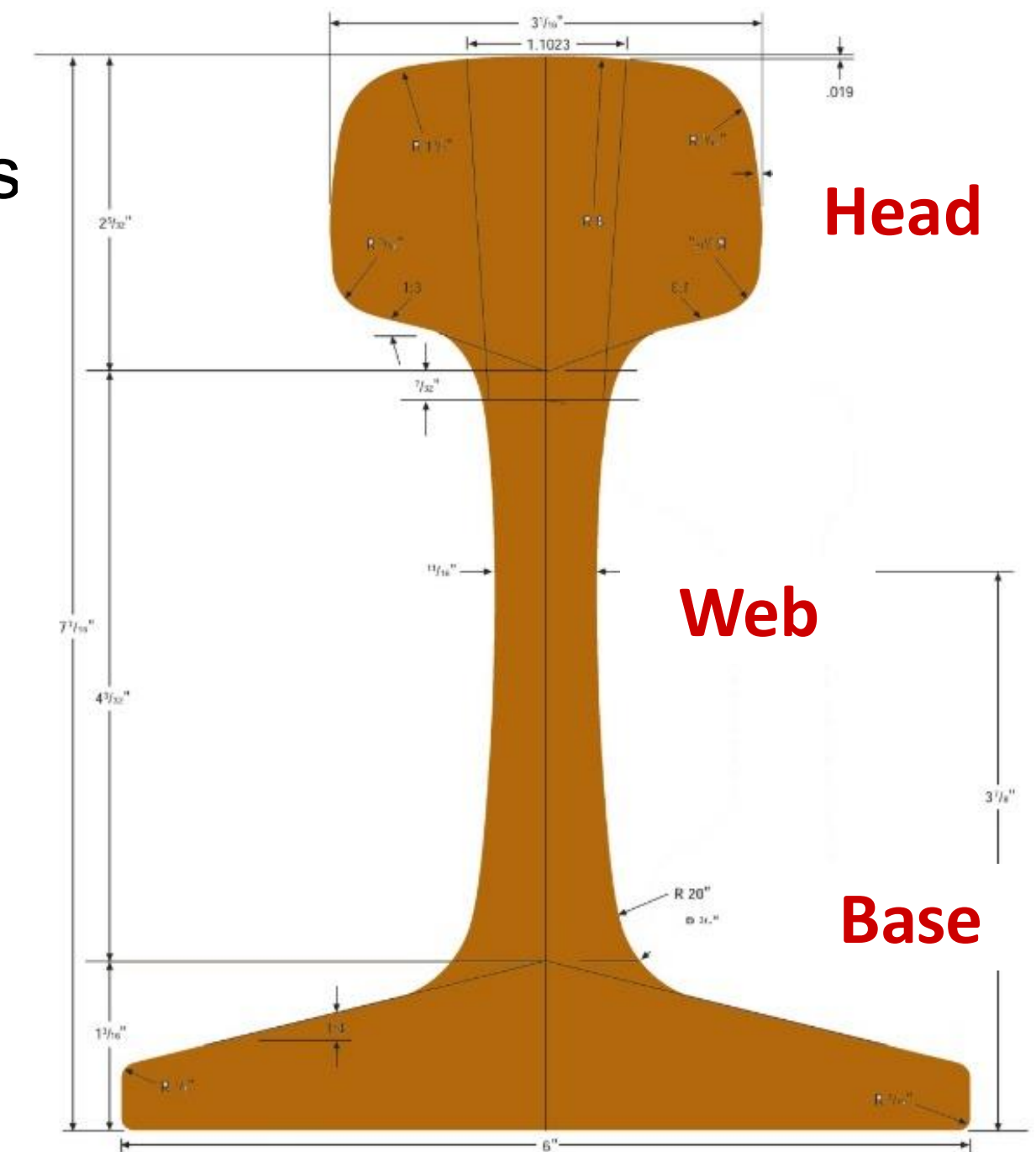
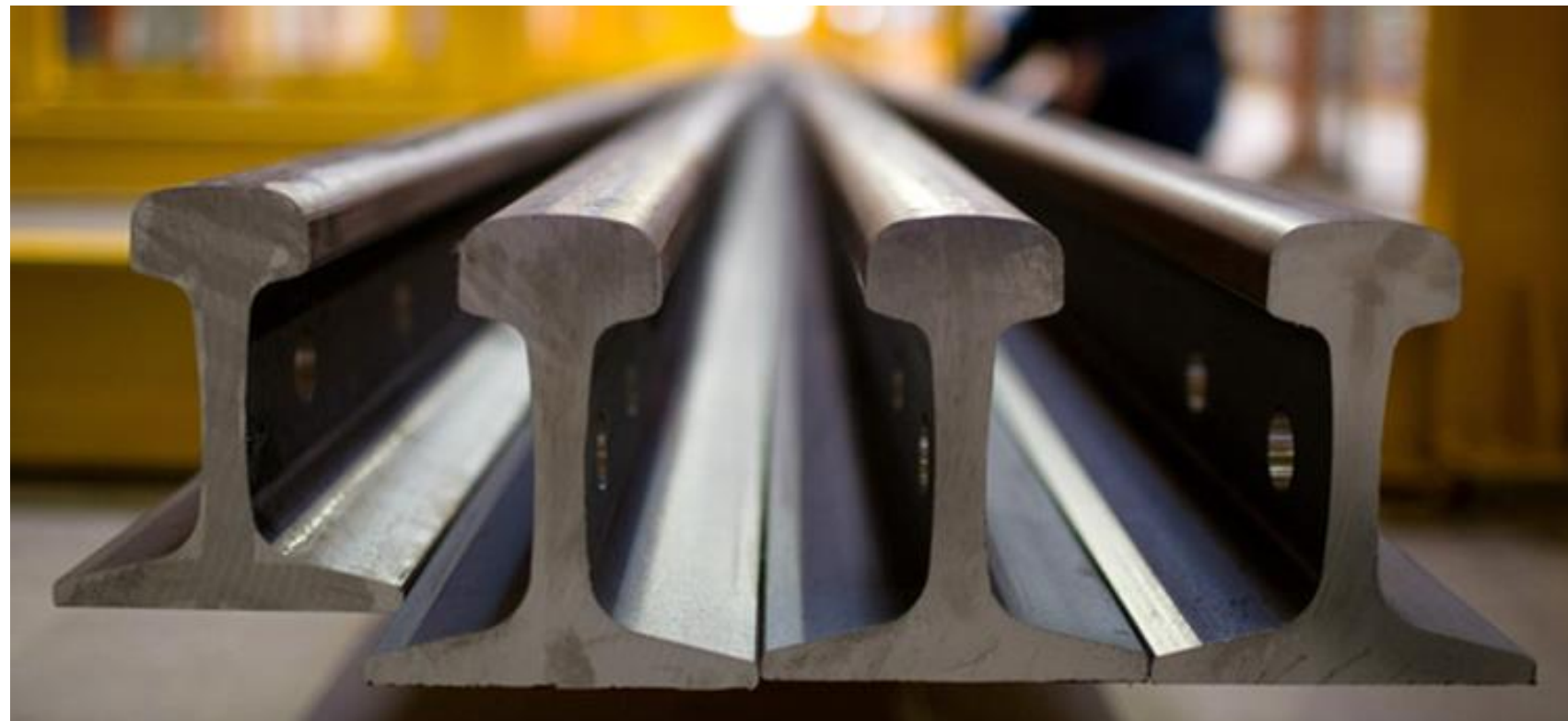
Track Components and Functions

Track Functions		Track Element or Component				
		Rail	Crossties	Fastening System	Ballast	Subballast
Primary	Support and Distribute Train Loads					
	Guide the vehicle					
	Provide adhesion at wheel-rail interface					
	Provide a smooth running surface					
	Facilitate drainage					
Secondary	Transmission of signal circuit					
	Broken rail detection					
	Path of ground return for traction power					

Credit: Progress Rail

Rail Basics and Function

- Rail is the single most valuable track asset owned by railroads
- Provides a smooth, low-friction, running surface for trains
- Wide, flat base distributes the load across several crossties and allows fasteners to be attached
- Combined with fasteners and crossties, the rail provides stable track gauge



Rail Size Measurement

US, Canada & UK: Rail size measured in *lbs per yard*

← one yard (36 inches) →

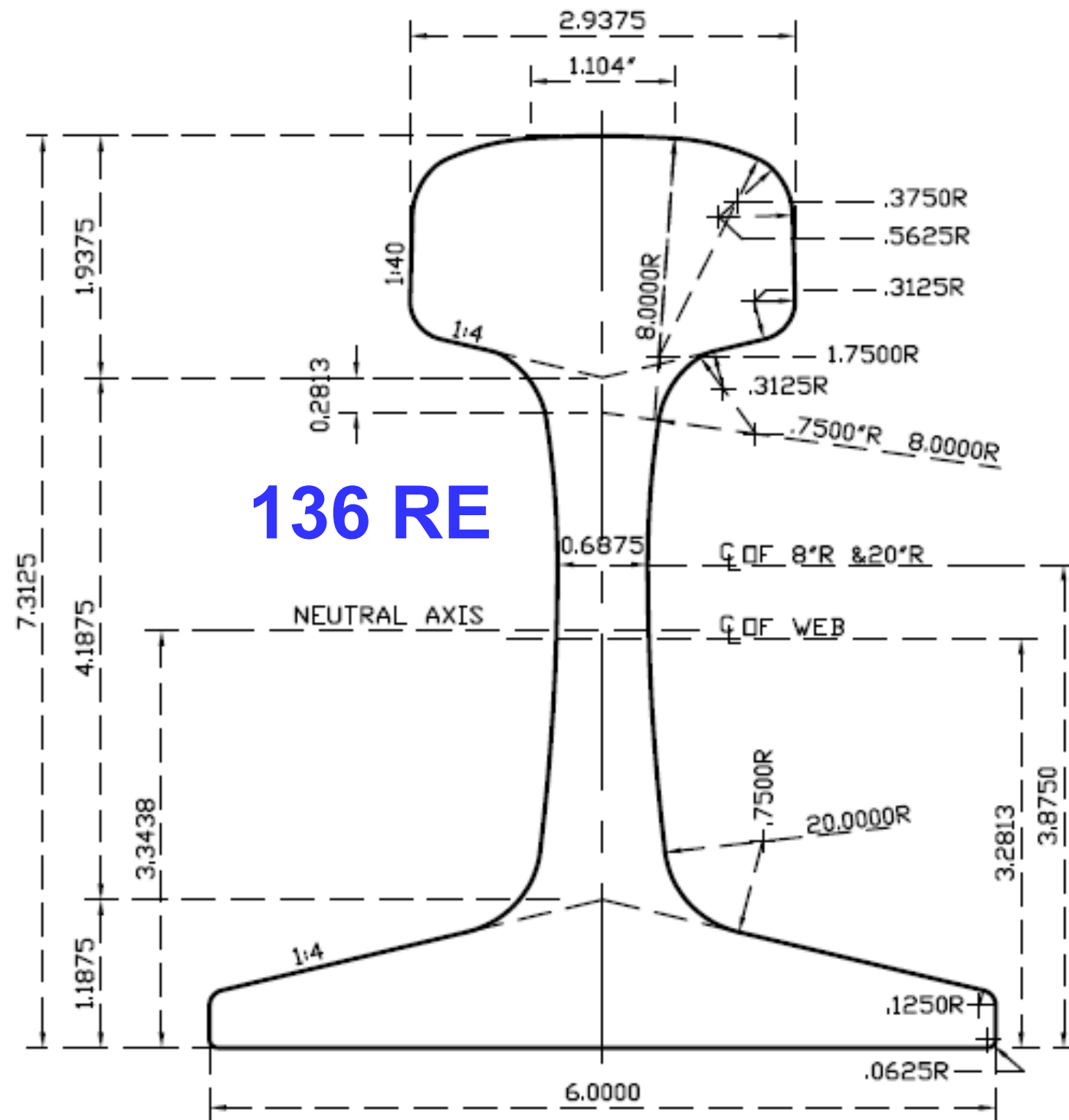


Elsewhere: Rail size measured in *kg per meter*

← one meter →

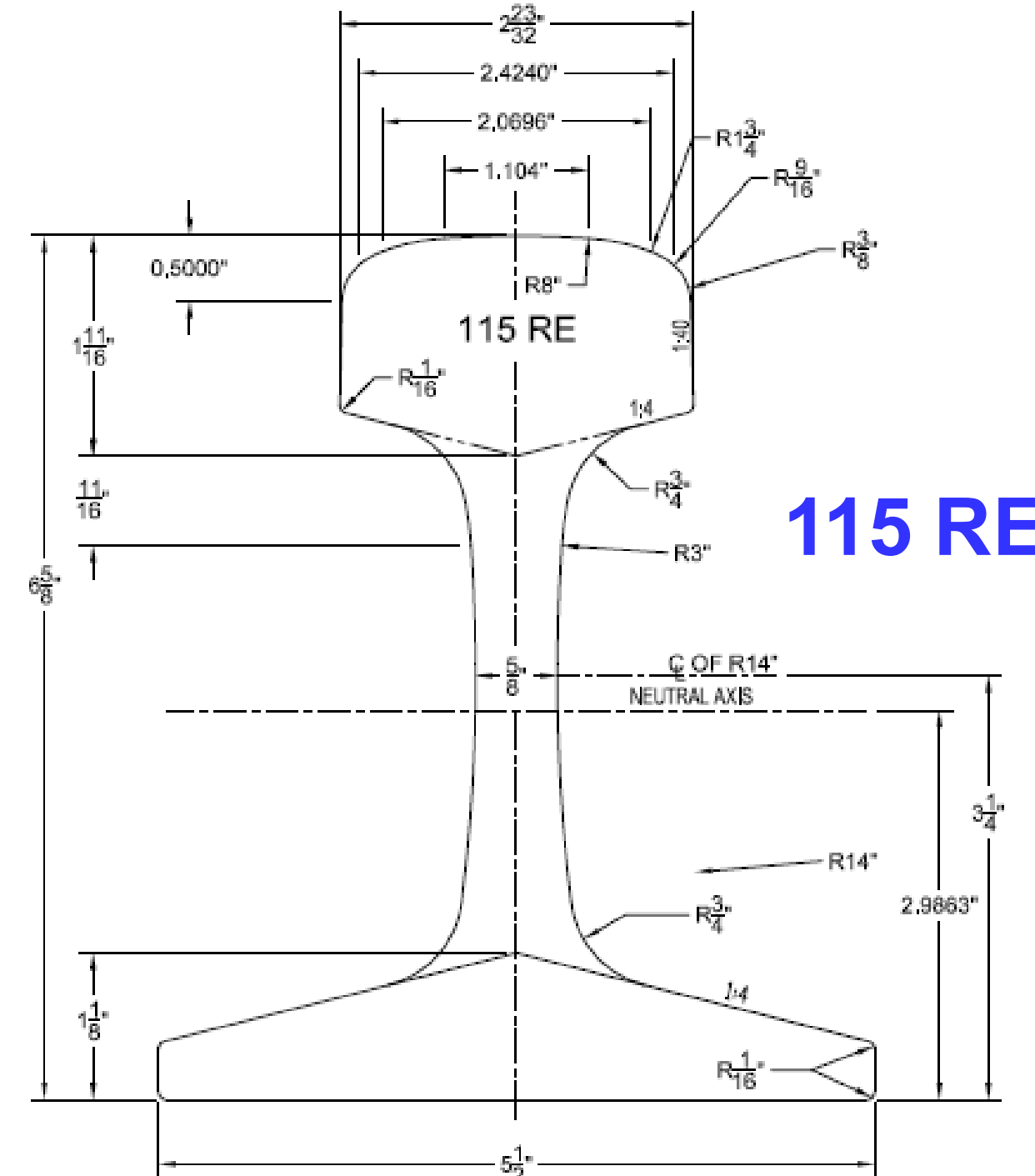


Common North American Rail Standards



136 RE

~60% of N.A. Rail Rolled



115 RE

~20% of N.A. Rail Rolled

Desirable Rail Qualities and Properties

High Wear Resistance

HARDNESS: the ability of a material to resist penetration, scratching, wear, abrasion and cutting

High Yield Point

DUCTILITY: the ability of a material to undergo relatively large plastic deformations before fracture

Resistance to Fatigue

TOUGHNESS: the ability of a material containing a crack to resist fracture

Summary of Rail Stresses & Effects

Type of Stress

Design / Maintenance Factors

Contact Stresses <i>(static and dynamic wheel loads at the wheel-rail interface)</i>	Bending Stresses <i>(static and dynamic wheel loads, non-uniform temperature changes)</i>	Axial Stresses <i>(uniform temperature changes, train forces and rail creep)</i>
1) Rail wear 2) Rail fatigue & shelling 3) Formation of plastic zone in contact region and rail corrugations	1) Selection of rail size 2) Rail section at poorly maintained joint which may plastically deform	1) Track buckling or rail pull-aparts 2) Selection of fastening system or distribution of rail anchors

And one more... residual stresses

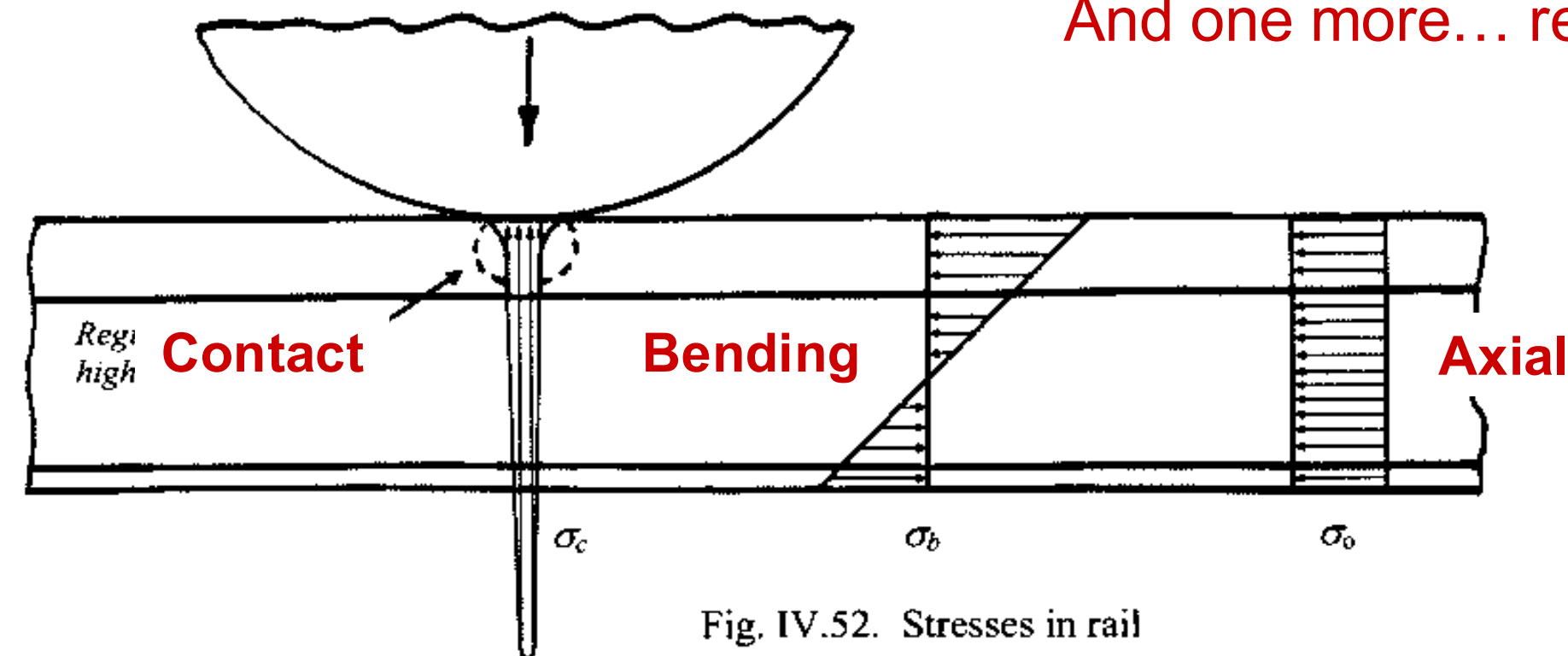
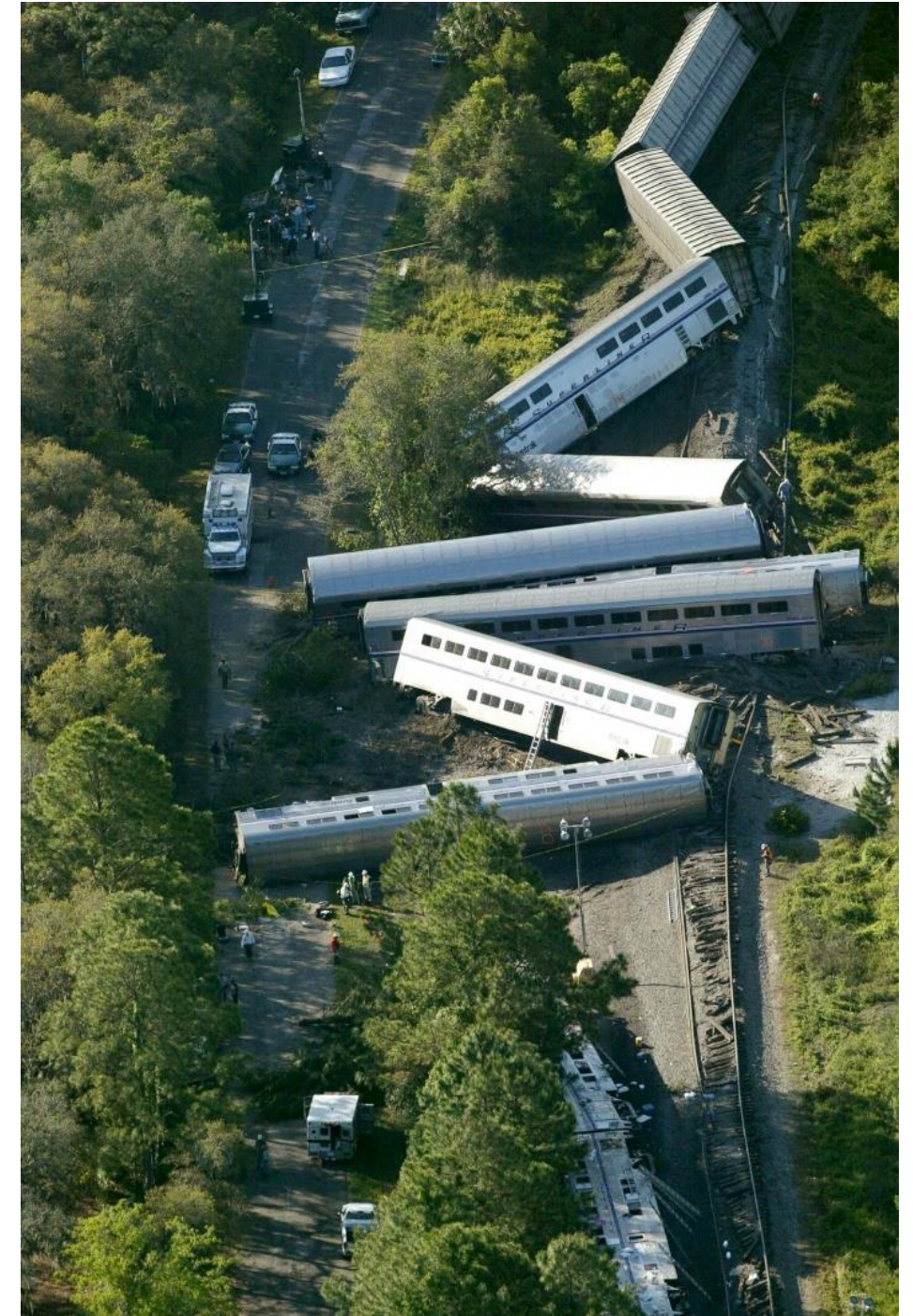


Fig. IV.52. Stresses in rail

Axial Stresses, Track Buckles, and Importance of Proper CWR Adjustment

- Large differences between **rail temperature** and **Rail Neutral Temperature (RNT)** generate high levels of tension or compression in the rail.
- High compressive forces that exceed the lateral and longitudinal strength of track can cause buckles due to:
 - Thermal loads alone (i.e., static buckling) or
 - Thermal and vehicle loads (i.e., dynamic buckling)
- Proper rail stress management for continuous welded rail (CWR) can prevent track buckles or rail breaks.
- Opportunities for adjustment of RNT:
 - Proper stressing at rail installation to set RNT
 - Rail adjustment / destressing and re-establishment of RNT

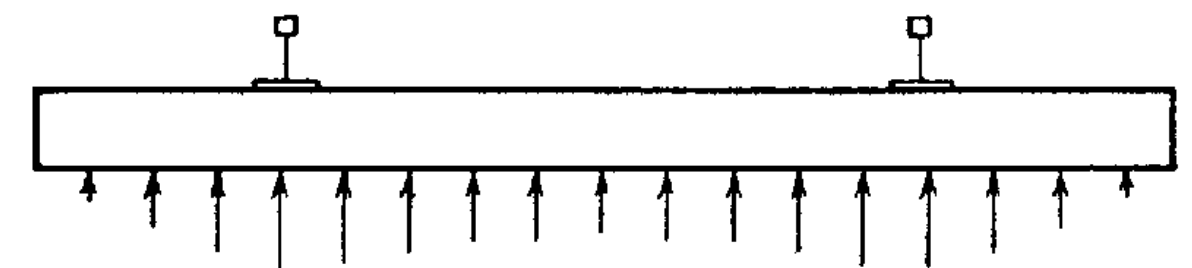
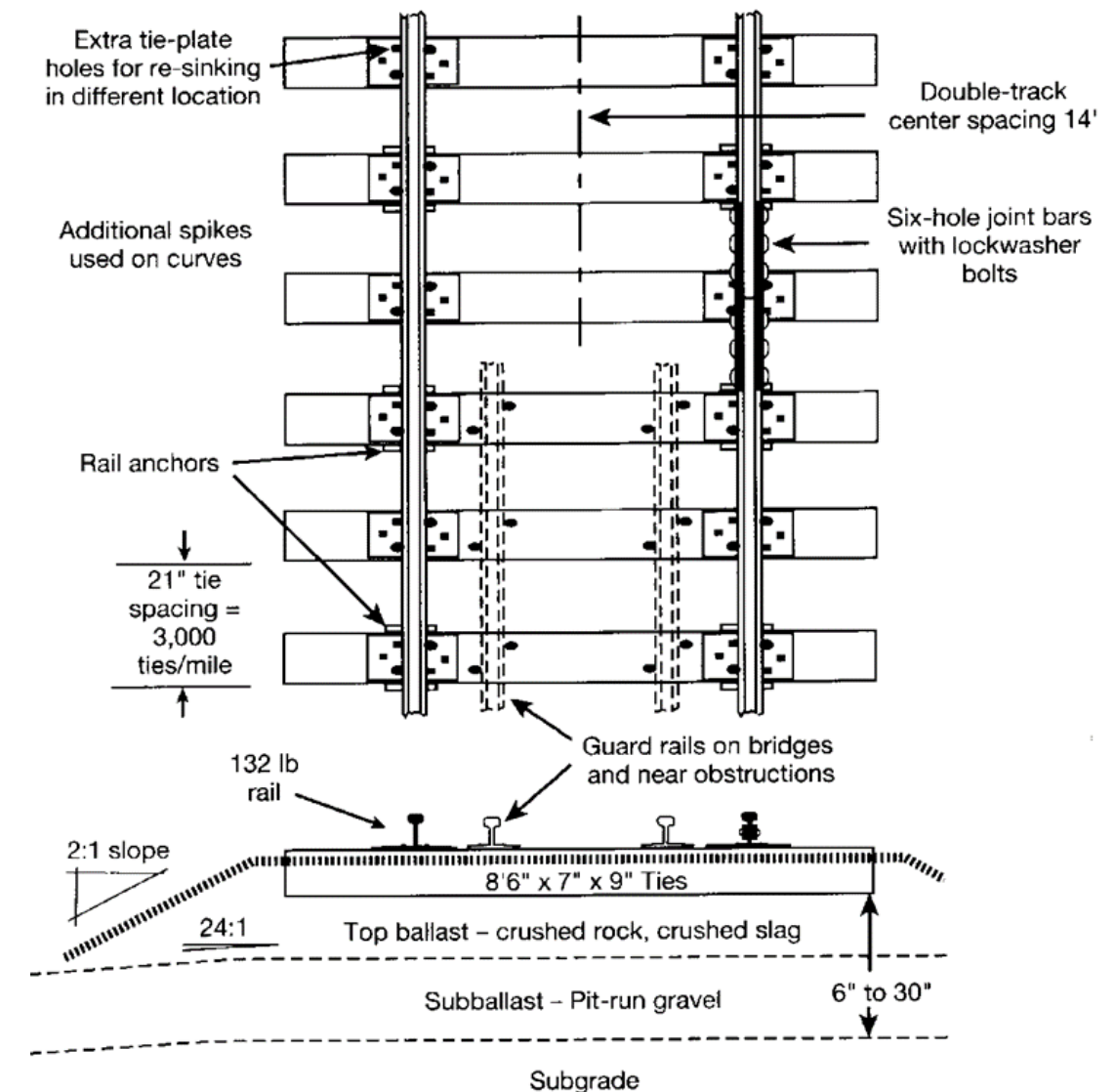


2002 Amtrak Auto Train Derailment on CSX
(<https://abcnews.go.com/US/timeline-major-train-crashes-us-2000/story?id=42450882>)

Crosstie Basics and Function

- Maintain gauge
- Distribute wheel loads from rails to ballast
- Anchor track against lateral, longitudinal, & vertical movement
- Spacing typically varies from 18" to 30":
 - 24 inch spacing $\approx$ 2,640 ties per mile **Common for Concrete**
 - 19.5 inch spacing $\approx$ 3,249 ties per mile **Common for Timber**
 - Crosstie spacing depends on tie material; tie size; ballast strength, joint spacing (supported or suspended) and track loading

Crosstie support is often uneven and highly variable in the field, changing bending demands



Types of Crossties

(700+Million in service in US)



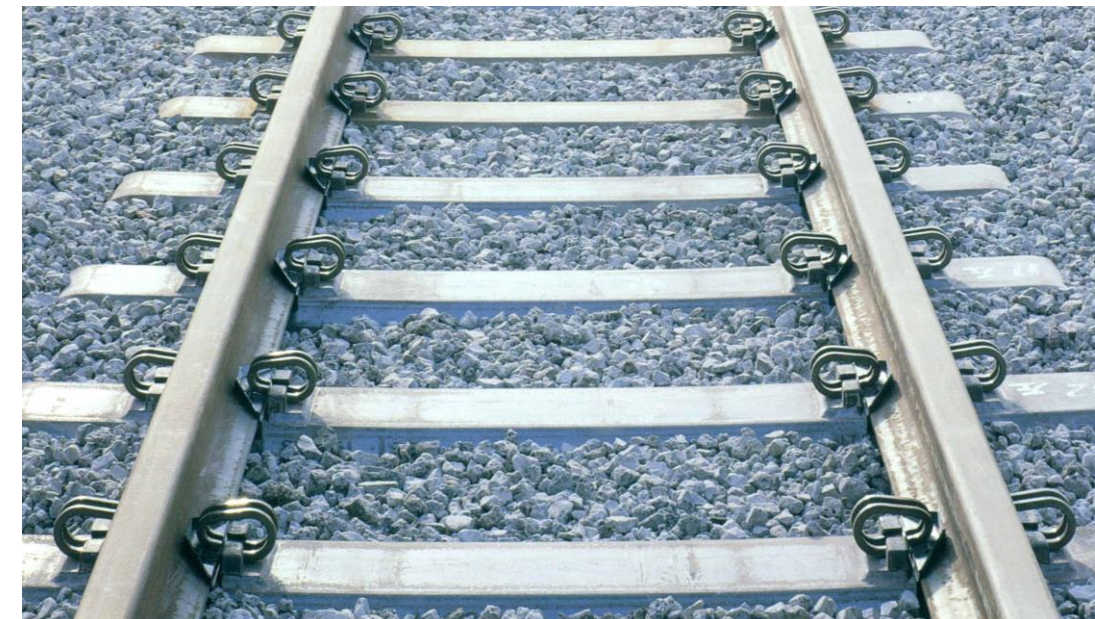
Timber (90-92%)



Concrete (6-8%)



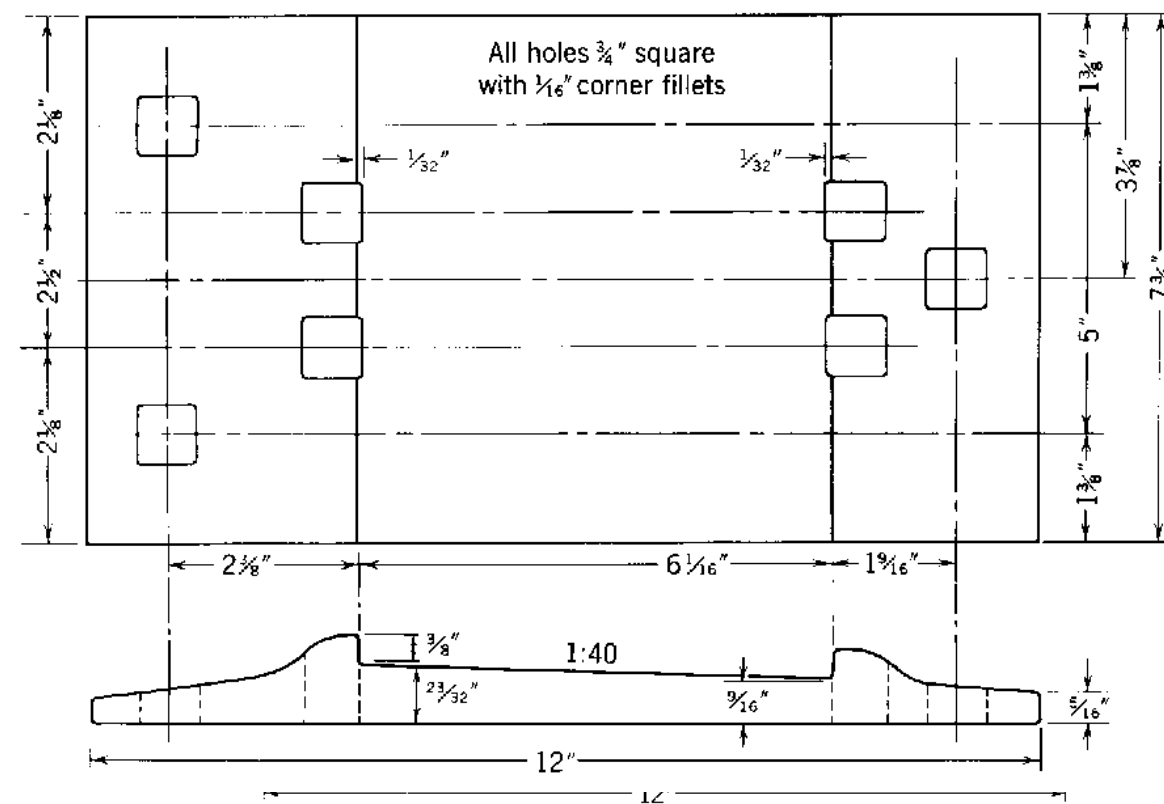
Composite (<1%)



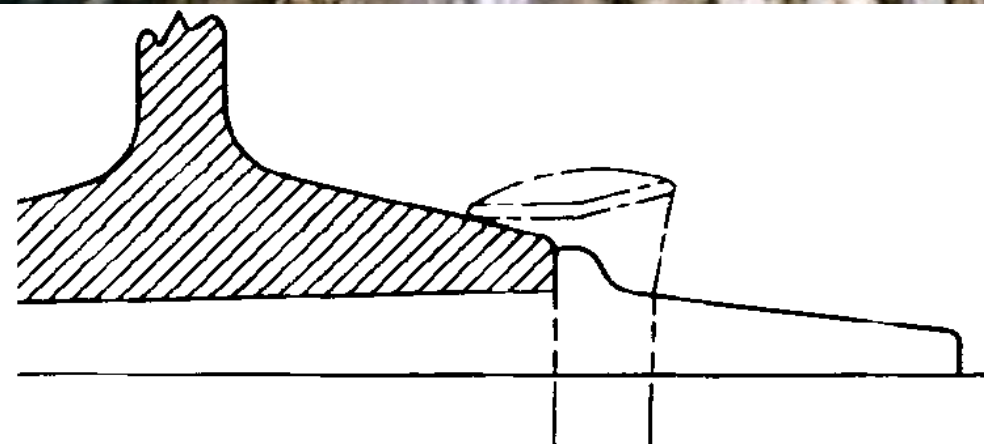
Steel (<1%)

Fasteners and Tie Plates

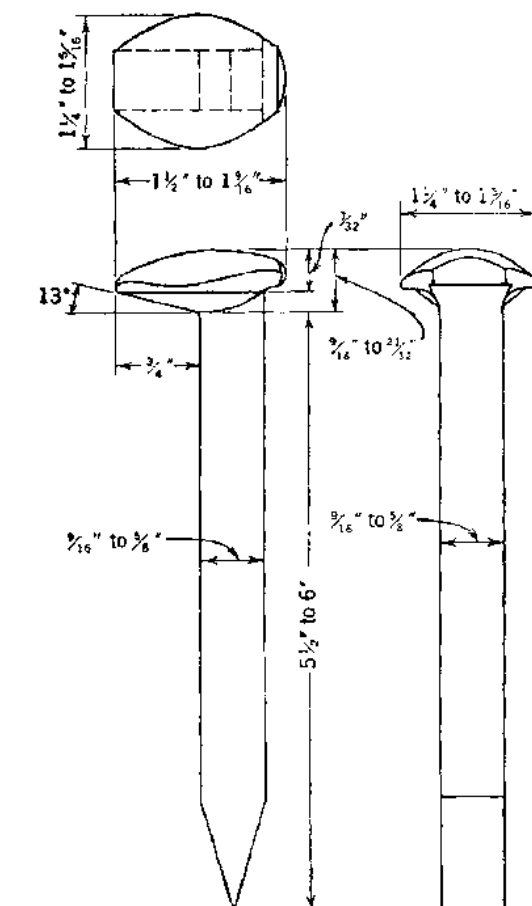
- Fasteners secure the rail to maintain track gage and provide torsional (in plane) track resistance.
- The tie plate supports the rail and distributes the load over a larger section of the tie surface.
 - The tie plate also provides lateral restraint to maintain gage and provides torsional resistance.
- Along with fasteners, crossties provide lateral, longitudinal and gage restraint and in turn further distribute the load into the ballast.



Tie Plate

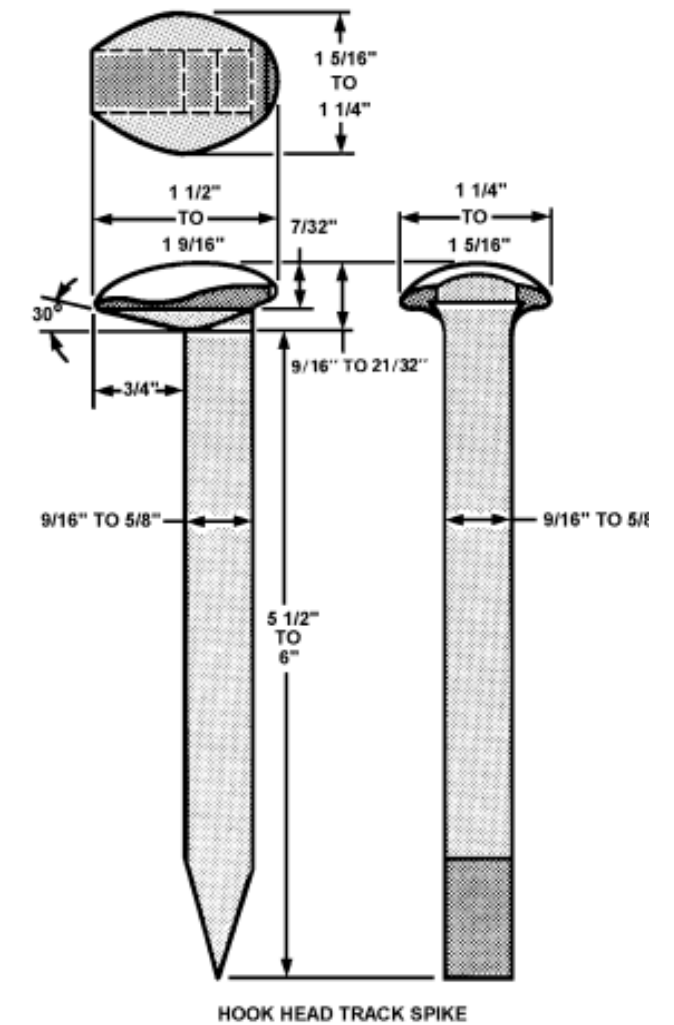


Standard “cut” spike



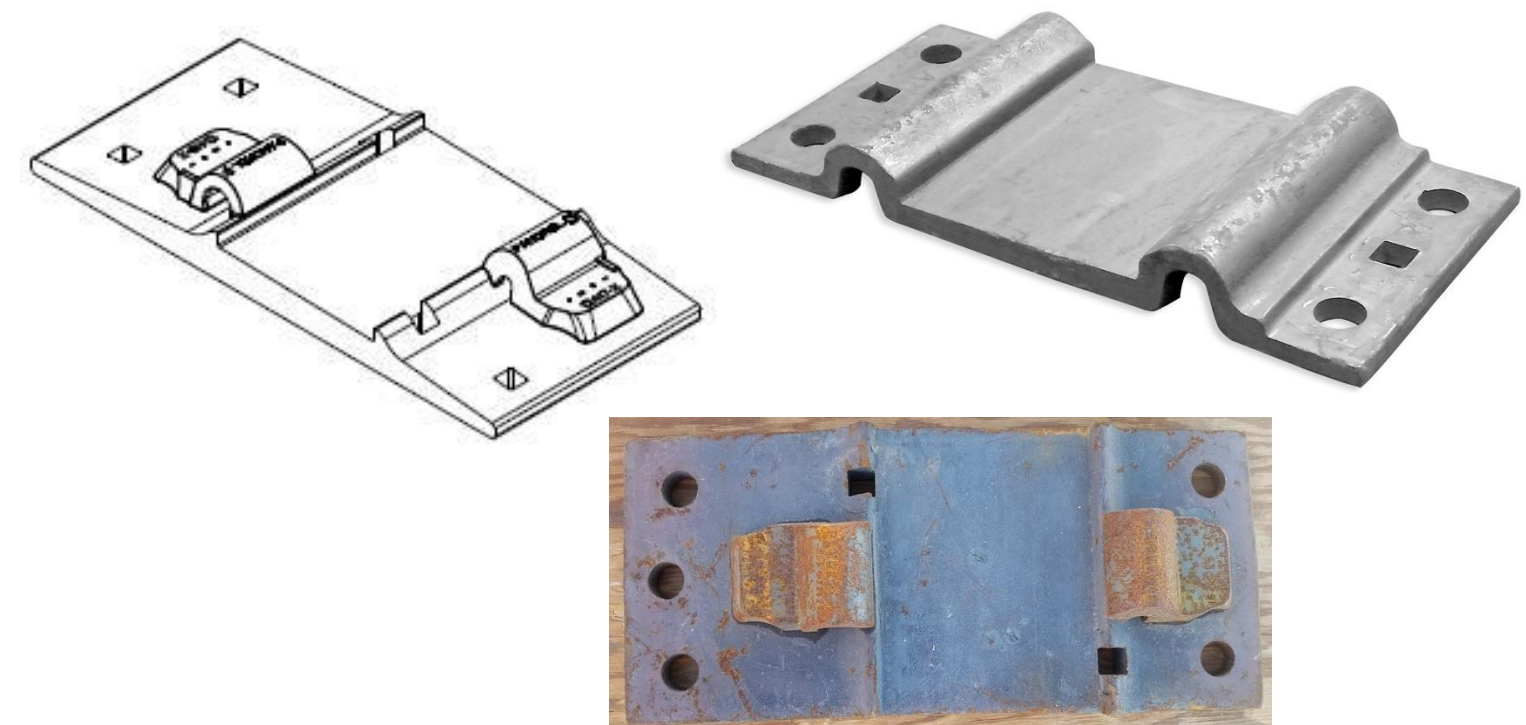
Common Timber Crosstie Fasteners

(~95% of timber track uses cut spikes and anchors)



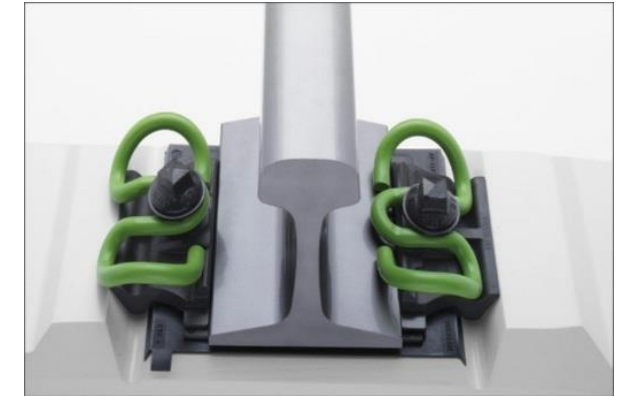
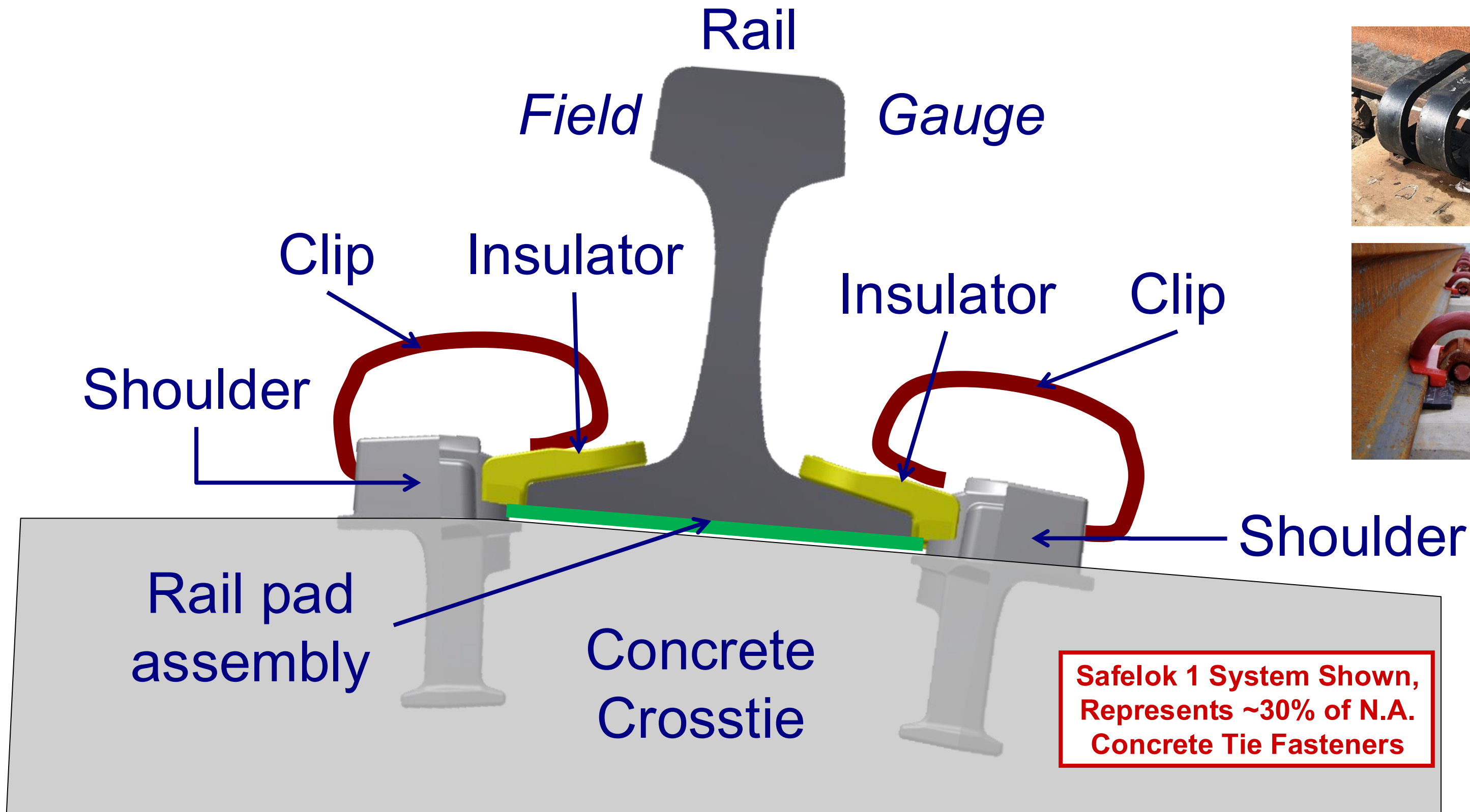
Elastic Fasteners and Premium Systems

(~5% of timber crosstie track)



Concrete Crosstie Fasteners

(Exclusive use of elastic fasteners)



Ballast & Sub-Ballast

- Ballast and sub-ballast are the final stages in load distribution prior to subgrade
- Distribute vertical loads and maintain the longitudinal and lateral stability of track
- Ballast and sub-ballast must provide adequate drainage
 - Ballast should be free draining and open graded (relatively uniform particle size)
 - Subballast should be densely graded, compacted, and crowned
- Ballast is subject to pulverization from loading as trains pass over, thereby generating fine particles that can “foul” the ballast and inhibit drainage
 - AREMA maximum ballast-crosstie pressures; 65 psi for timber crossties, 85 psi for concrete crossties



Ballast

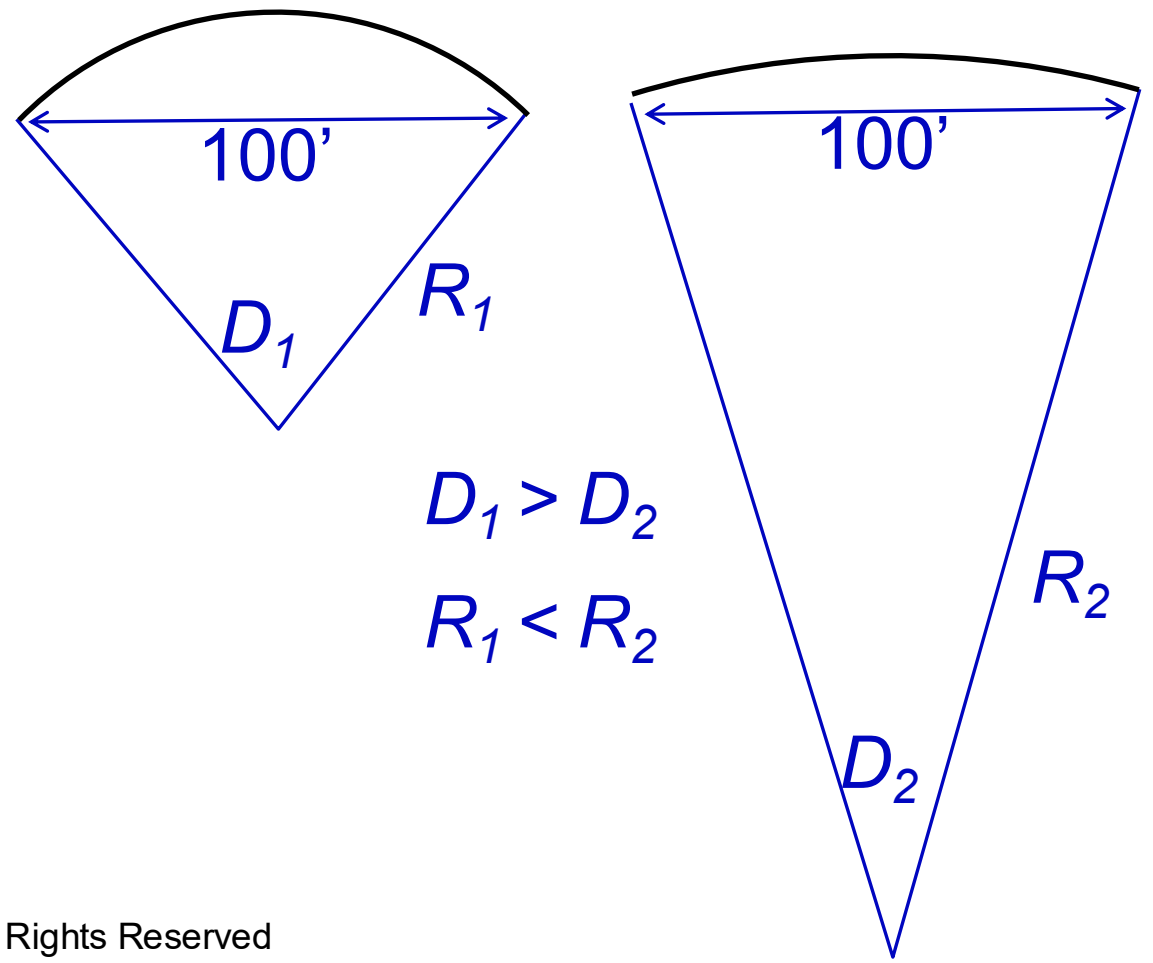
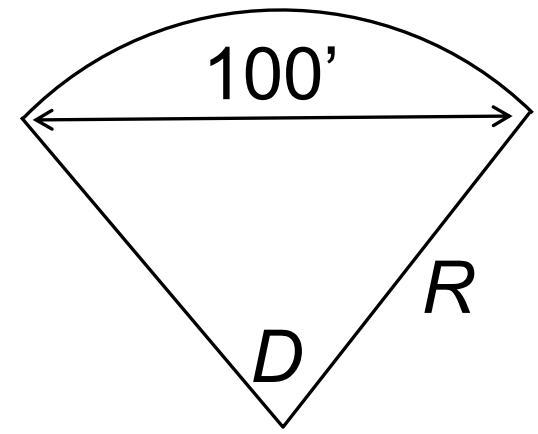


Subballast

Horizontal Curvature, Superelevation, Spirals, and Track Design

Horizontal Curvature: Simple Circular Curves

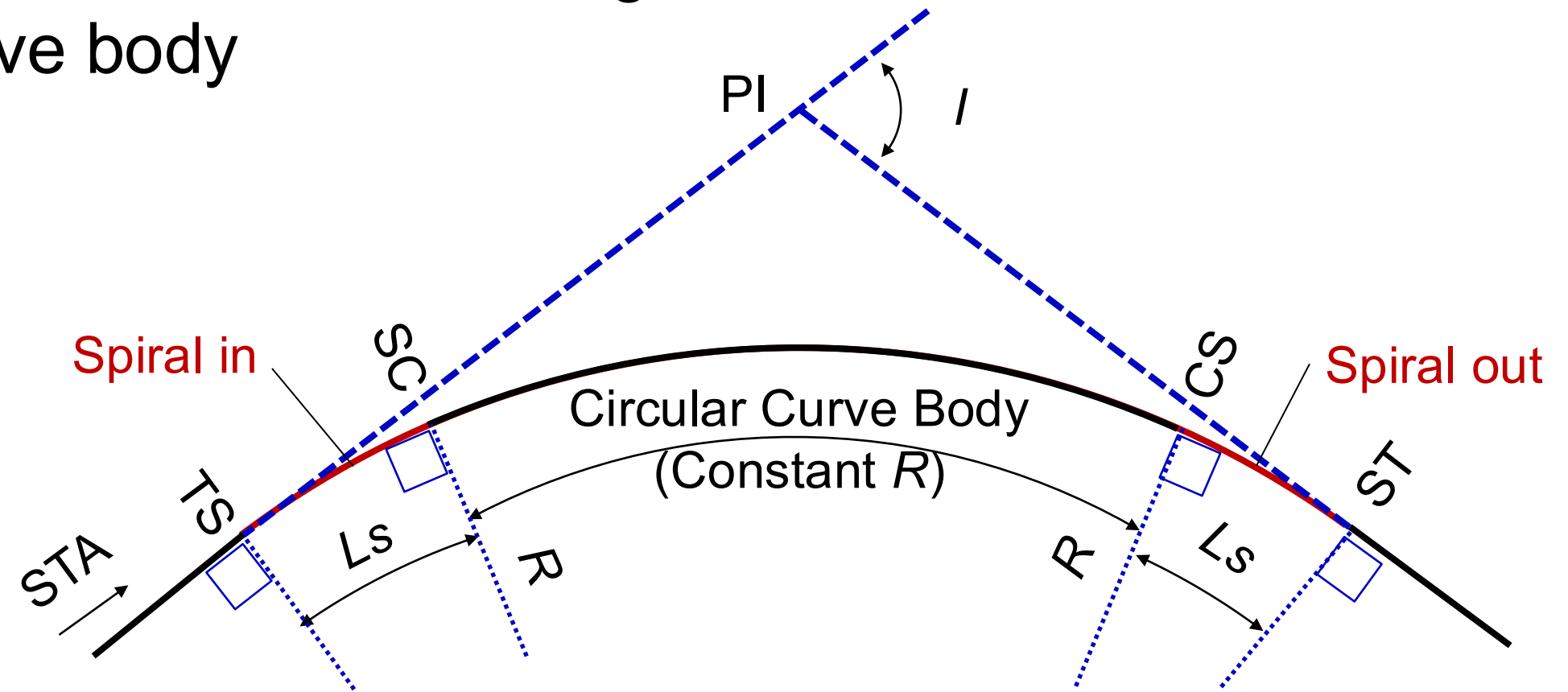
- Railways use the **chord** definition for curves
 - Angle measured along the length of a section of curve subtended by a 100' chord
- Lower degree of curve = larger radius = more gradual curve:



Typical Curve Degrees (Applications)	Degree of Curve
Mainline (desirable)	$< 2^\circ$
Mainline (typical)	$< 4^\circ$
Secondary branchline (typical)	$< 6^\circ$
Unit train facility (maximum)	$< 7.5^\circ$
Industrial or yard track (desirable)	$< 9.5^\circ$
Industrial or yard track (maximum)	$< 12.5^\circ$

Horizontal Curvature: Spiral Curves

- Spiral transitions or “clothoids” are used to gradually “steer” railcars into curves and develop superelevation (*unlike highways, we need to design “steering” into the track*)
- Spiral starts at tangent with **infinite radius** and then tightens until it matches the **circular curve radius** at the start of the curve body

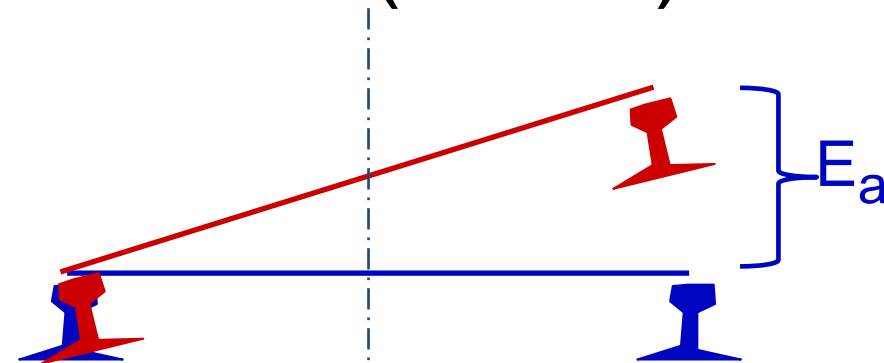


- Spiral length (L_s) is a function of:
 - Design speed, actual superelevation (E_a), and superelevation runoff rate
 - Railcar length (determines minimum spiral length to prevent car warp)
- Each railroad has its own process of calculating spiral length

Superelevated Spiral Curves

(Combining horizontal curvature change and superelevation)

- **Most** railway curves use superelevation and spiral transitions
- Actual superelevation “ E_a ” refers to the elevation difference (in inches) between “high rail” (outside) and “low rail” (inside) on a curve



- Typical maximum value for E_a on freight railroads is 5 inches
- Track is rotated about the low rail in curves (as opposed to centerline for highways)
 - Unlike highways which rotate about the centerline and the inside edge drops down
- Superelevation is established over a spiral transition between the tangent and the circular portion of the curve
- Circular portion of the curve is called the “curve body”

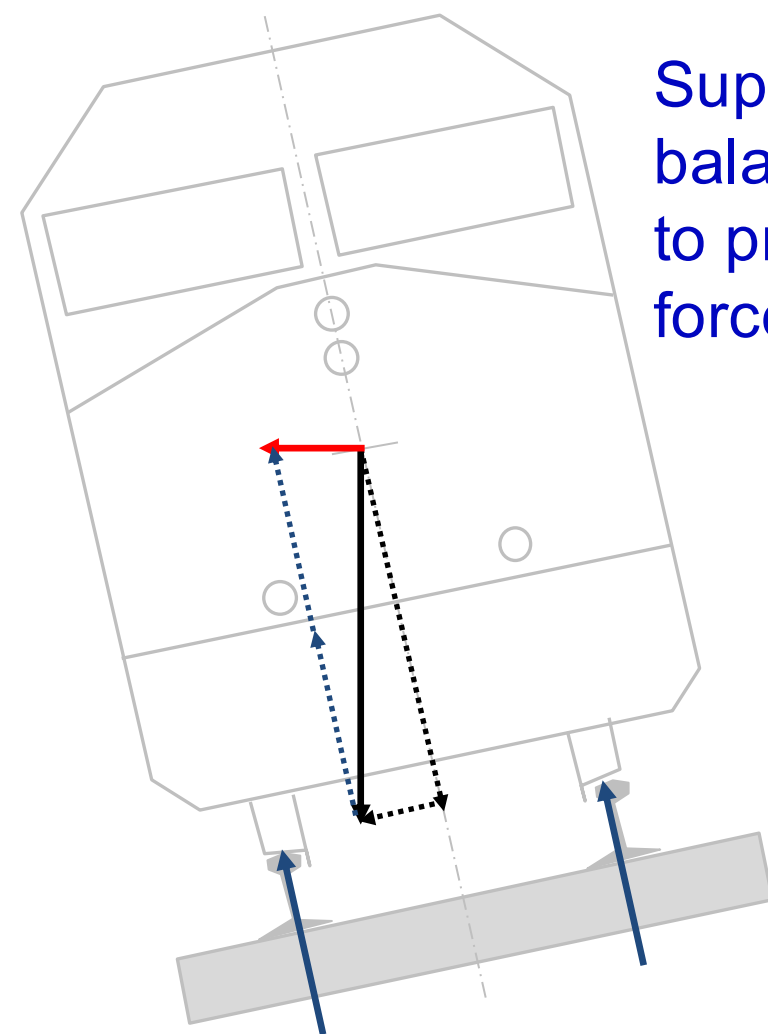
Balanced vs. Actual Superelevation

(Cant Deficiency)

- E_a is a function of speed (V in mph), degree of curve (D), and E_u
- E_u is allowable unbalance (also referred to as *cant deficiency*)
 - typically 1" to 2" for freight operations, depending on the railway

Balanced Condition

Superelevation exactly balances curving forces to produce equal wheel forces on each rail



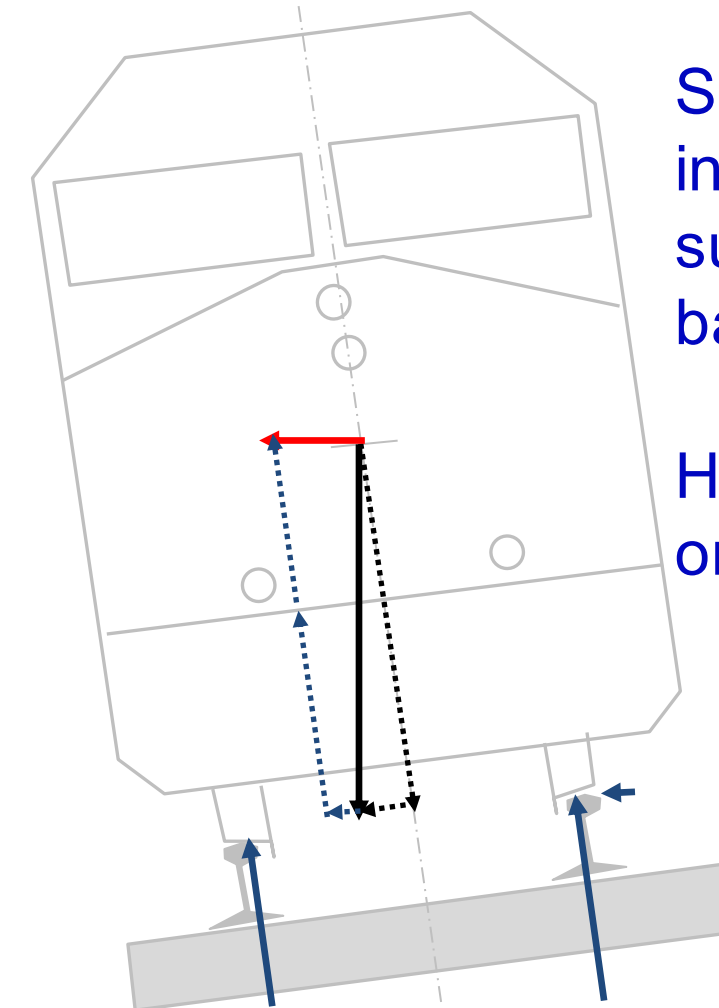
} E_b

$$E_b = 0.0007V^2D$$

Underbalance Condition

Same curve speed but install less actual superelevation than balanced condition

Higher forces on outside or "high rail"



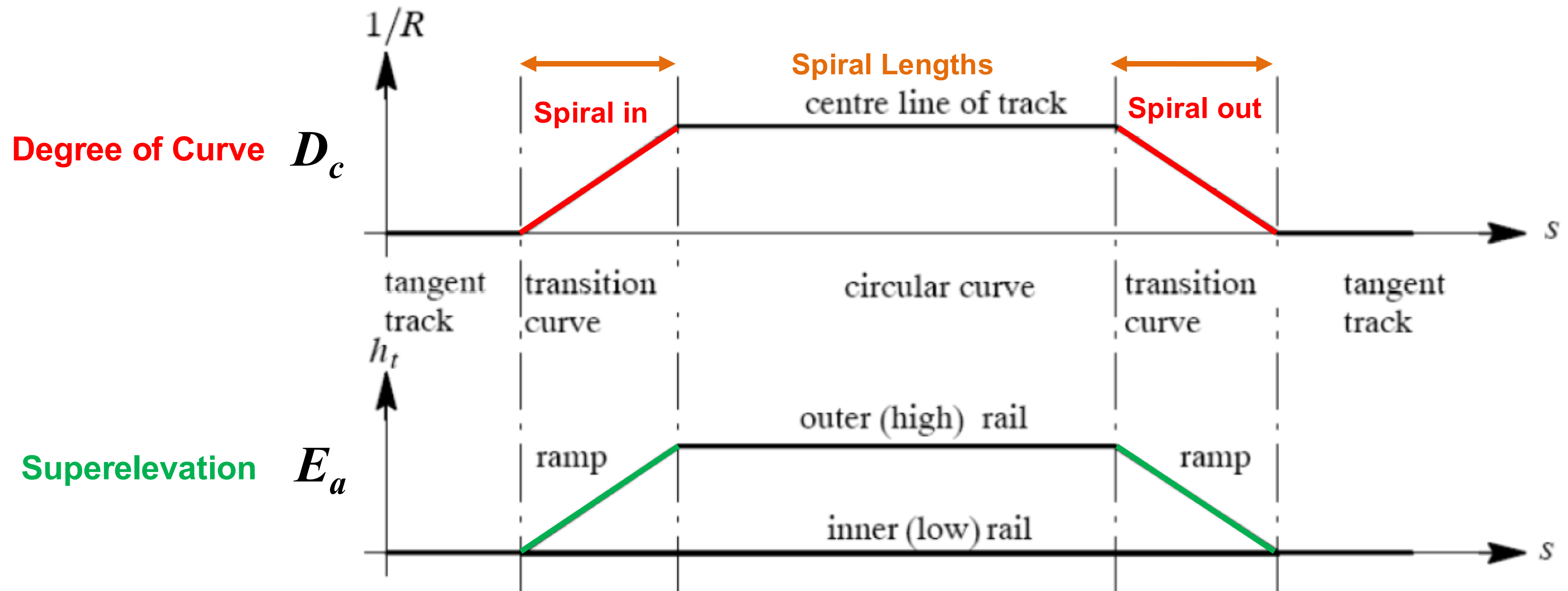
} $E_a = E_b - E_u$

$$E_a = 0.0007V^2D - E_u$$

Resultant Force →
Wheel/Rail Forces →
Gravitational Force →

Curve Transitions – Spirals & Superelevation

In most applications, curvature and superelevation change **linearly** with distance along spiral



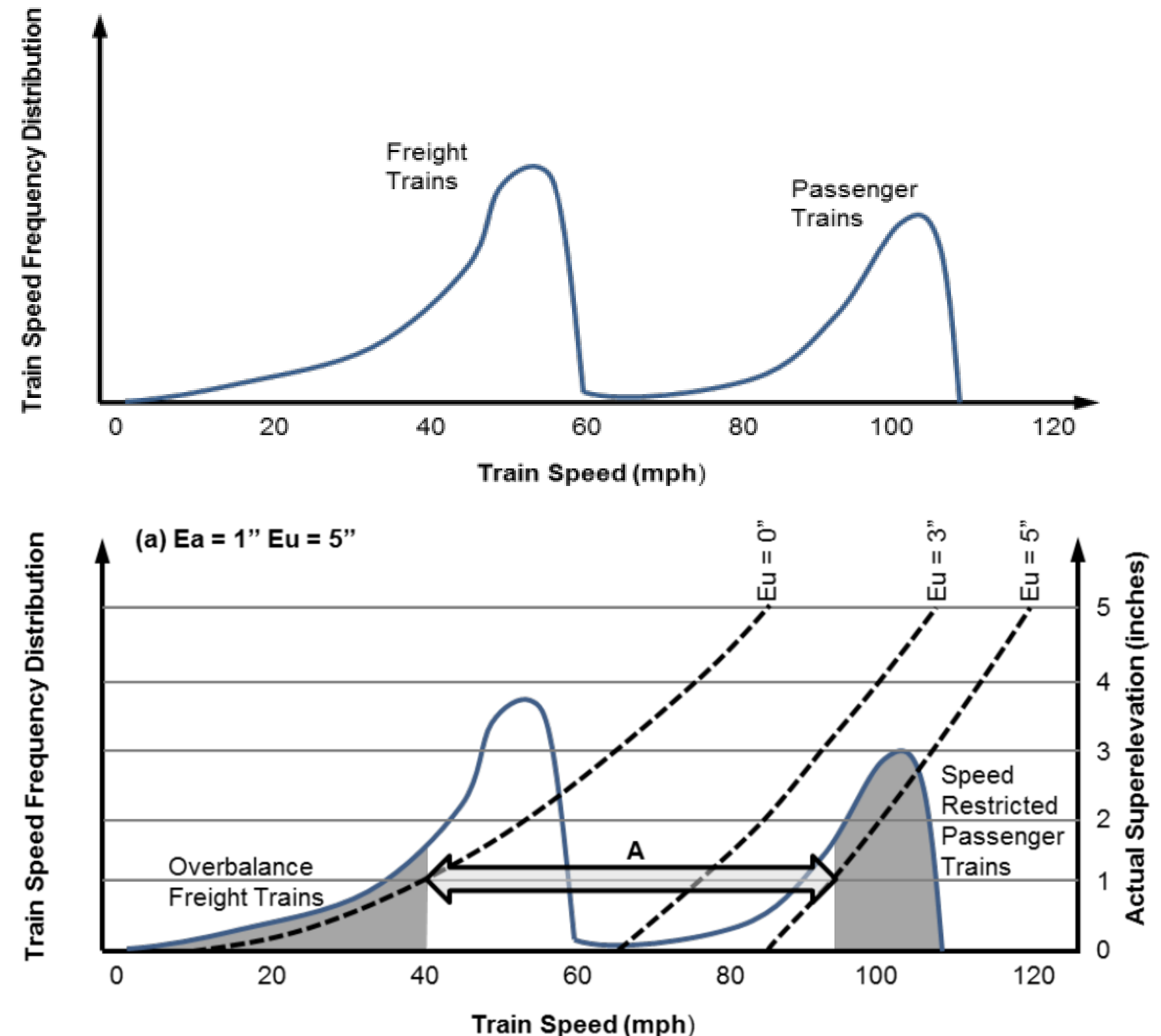
Superelevated Spiral Curve Example

- Example Problem (related to case study):
 - Timetable speed: 70 mph
 - Allowable unbalance (E_u) for freight trains: 2 inches
 - Degree of curve: 1.85 degrees ($1^\circ 51'$)
- First, calculate **balanced superelevation (E_b)**
 $E_b = 0.0007V^2D = (0.0007)(70)(70)(1.85) = 6.346$ inches (6-3/8" rounded)
- Second calculate (design) **actual superelevation (E_a)**
 $E_a = 0.0007V^2D - E_u = (0.0007)(70)(70)(1.9) - 2 = 4.346$ inches (4-3/8" rounded)
- Look up standard **superelevation runoff rate** for design speed
 - For 70 mph, runoff rate is 82 feet per inch (see table to right)
- Multiply runoff rate by E_a to determine **spiral length (L_s)**
 $L_s = E_a \times \text{runoff rate} = (4.375)(82) = 358.75$ feet
Round L_s up to the nearest 10-foot increment, $L_s = 360$ feet

MAXIMUM SPEED FOR CURVES IN MPH	PREFERRED DISTANCE REQUIRED FOR EACH INCH CHANGE IN THE SUPERELEVATION
COLUMN 1	COLUMN 2
20	44
25	44
30	44
35	44
40	47
45	53
50	59
55	65
60	70
65	76
70	82
75	88
80	94
85	100
90	106

Heterogeneity in Operations and Superelevation

- Speeds are likely to vary due to:
 - Different types of trains
 - Passenger vs. Freight
 - Intermodal vs. Manifest
 - Loaded Unit vs. Empty Unit
 - Train direction – ascending or descending grades
- Combined with a fixed E_u and E_a , the demands on a given curve vary greatly depending on the train speed
- **Very useful reference:** Dick, C.T. and C. Ruppert. 2019. *Mixed Freight and Higher-Speed Passenger Trains: Framework for Superelevation Design*. FRA Technical Research Report: https://railroads.dot.gov/sites/fra.dot.gov/files/fra_net/19085/Superelevation.pdf

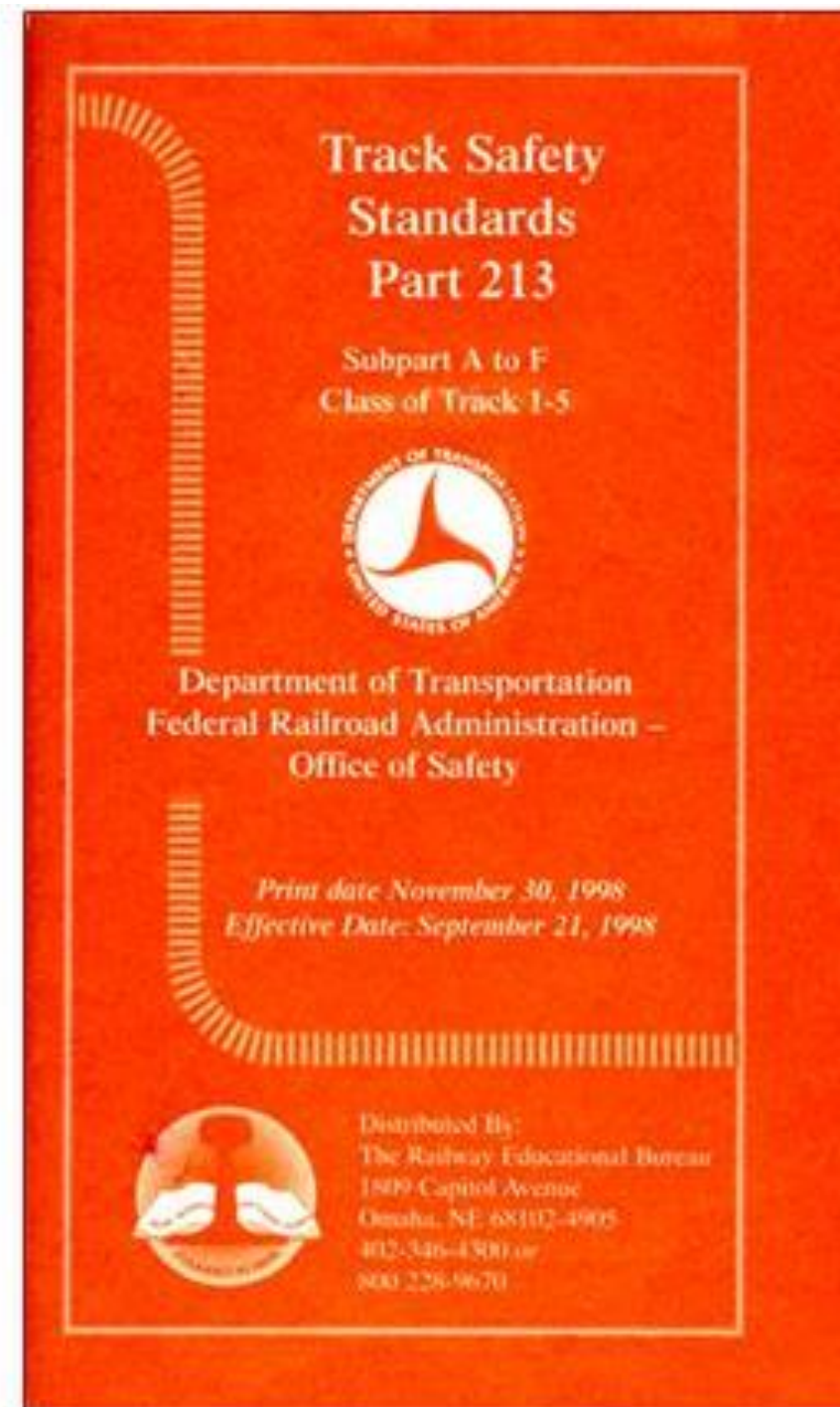


Track Standards and Regulations

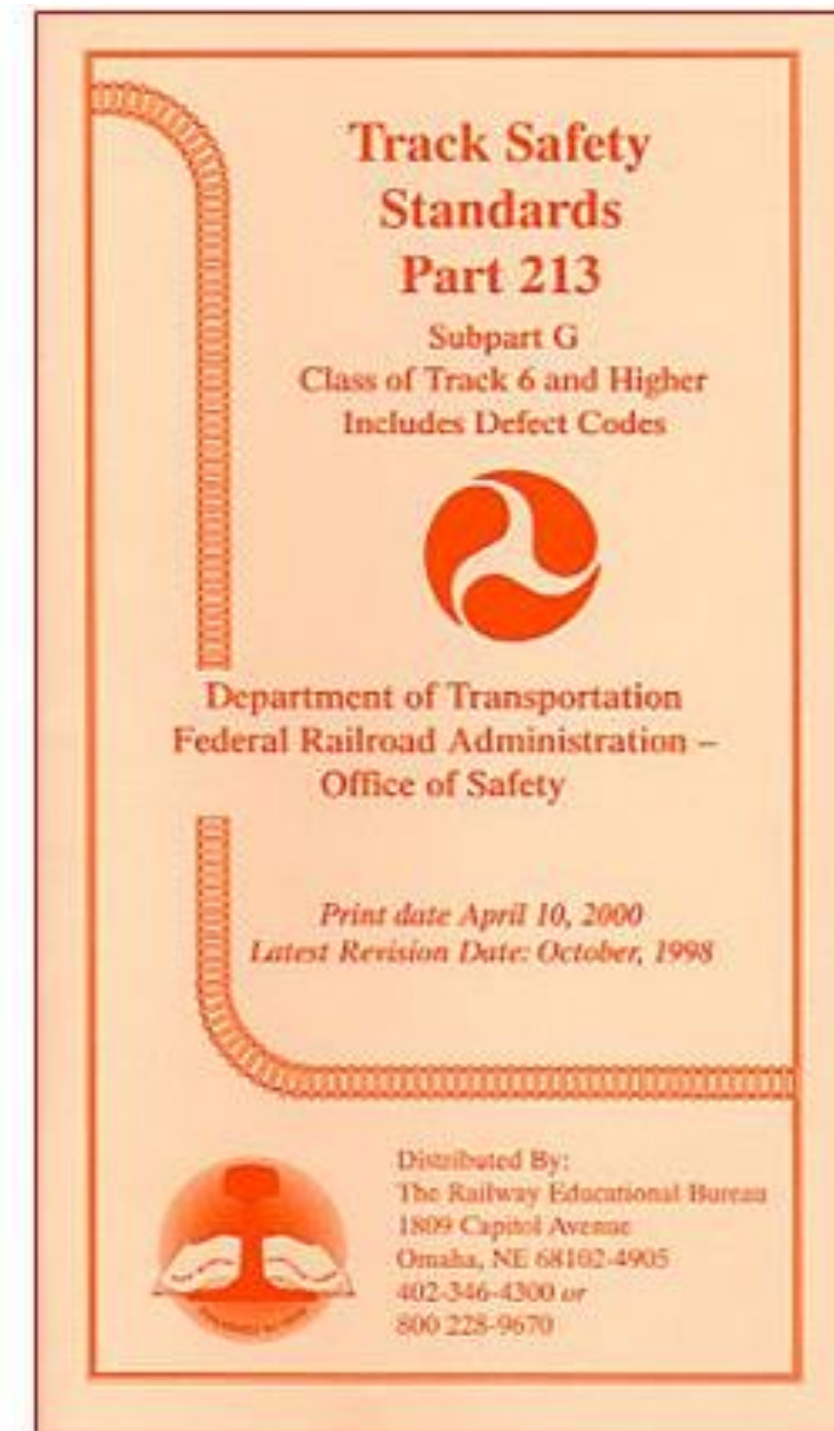
Track Standards & Specifications

- Design Standards & Recommended Practices (Ideal)
 - desired structural capacity and geometric configuration
- Construction Standards & Specifications (Tolerances)
 - desire is to build to zero tolerance from the plans & specs
 - in practice allowable tolerances are set to account for manufacturing limits, construction methods, etc.
- Maintenance Standards (Thresholds)
 - used to trigger maintenance or reconstruction actions
- Safety Standards (Limits)
 - limits, once passed, that require the immediate repair of track, speed restrictions or removal from service

FRA Track Safety Standards



FRA Track Classes 1-5



FRA Track Classes 6-9

FRA Compliance Manual

Track and Rail and Infrastructure Integrity Compliance Manual

**Volume II Track Safety Standards
Chapter 1 Track Safety Standards
Classes 1 Through 5**

January 2017

Office of Railroad Safety



U.S. Department of Transportation
Federal Railroad Administration

Federal Railroad Administration



Track and Rail and Infrastructure Integrity Compliance Manual

**Volume II Track Safety Standards
Chapter 2 Track Safety Standards
Classes 6 through 9**

January 2017

FRA Track Class and Train Speed

- Maximum speeds are defined by FRA Track Class:

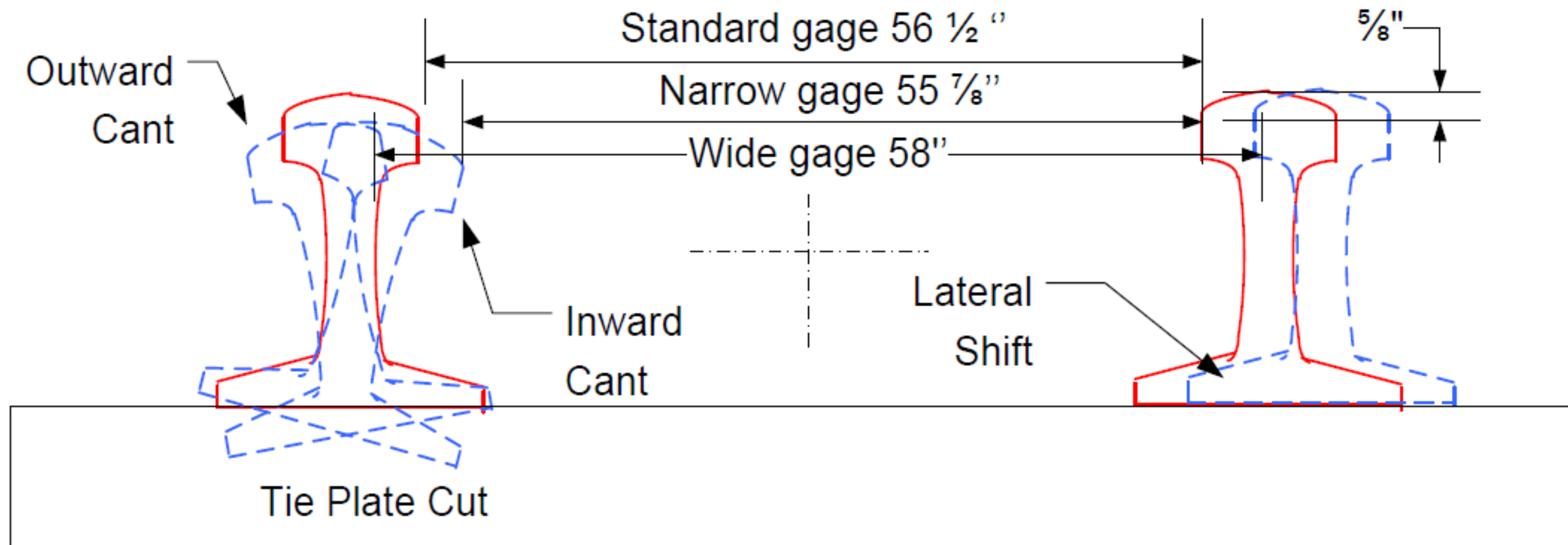
FRA Track Class	Freight Train Speed	Passenger Train Speed
1	10	15
2	25	30
3	40	60
4	60	80
5	80	90

- Track which does not comply with the standards for its posted Track Class **MUST** be **fixed, protected, or removed from service** as soon as the condition is found (or **operated at a lower Track Class / speed**)

Track Gage

§ 213.53 Gage

53(a) Gage is measured between the heads of the rails at right angles to the rails in a plane five-eighths of an inch below the top of the rail head.



Track Gage

§ 213.53 Gage

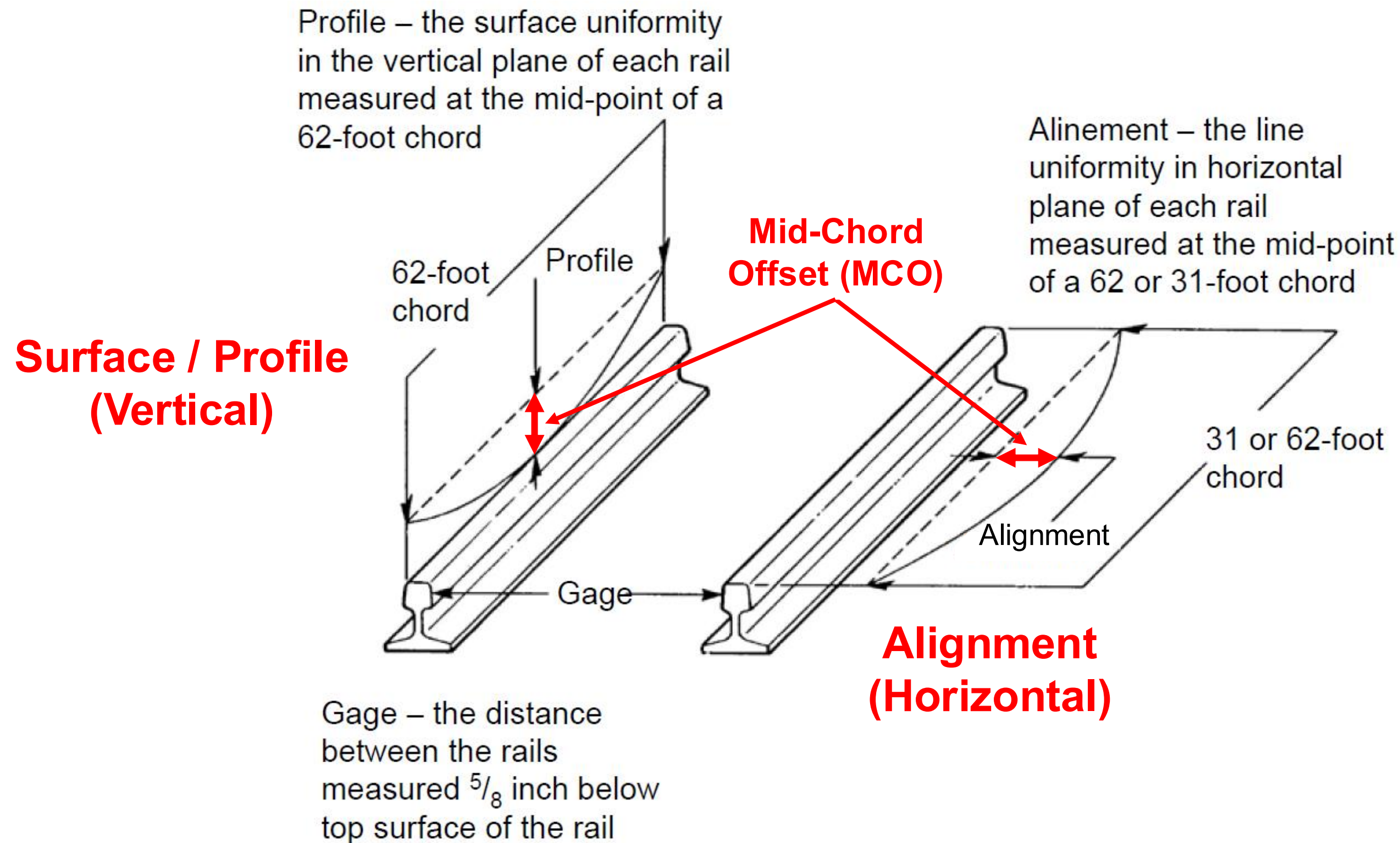
53(a) Gage is measured between the heads of the rails at right angles to the rails in a plane five-eighths of an inch below the top of the rail head.

<i>Class of Track</i>	<i>The gage must be at least -</i>	<i>But not more than -</i>
<i>Excepted track</i>	N/A	4'10¼"
<i>Class 1 track</i>	4' 8"	4' 10"
<i>Class 2 and 3 track</i>	4' 8"	4' 9¾"
<i>Class 4 and 5 track</i>	4' 8"	4' 9½"



(1/2" Narrow
to 1" Wide)

Track Geometry Measurements



Alignment Measurement (Stringlining a Curve)

Mid-Chord Offset
(MCO)

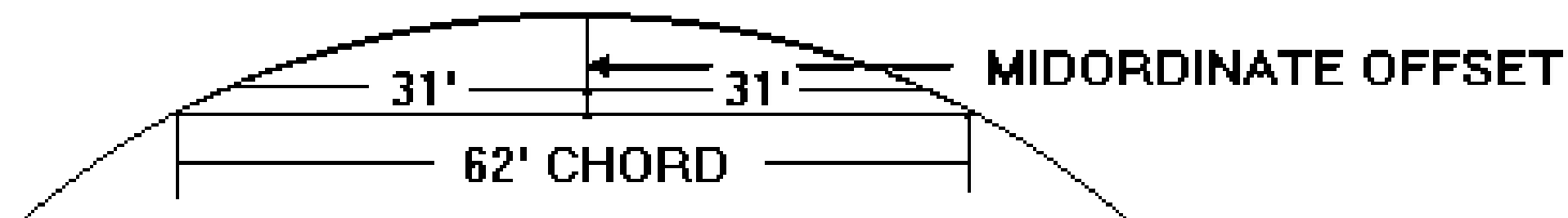


Track Alignment

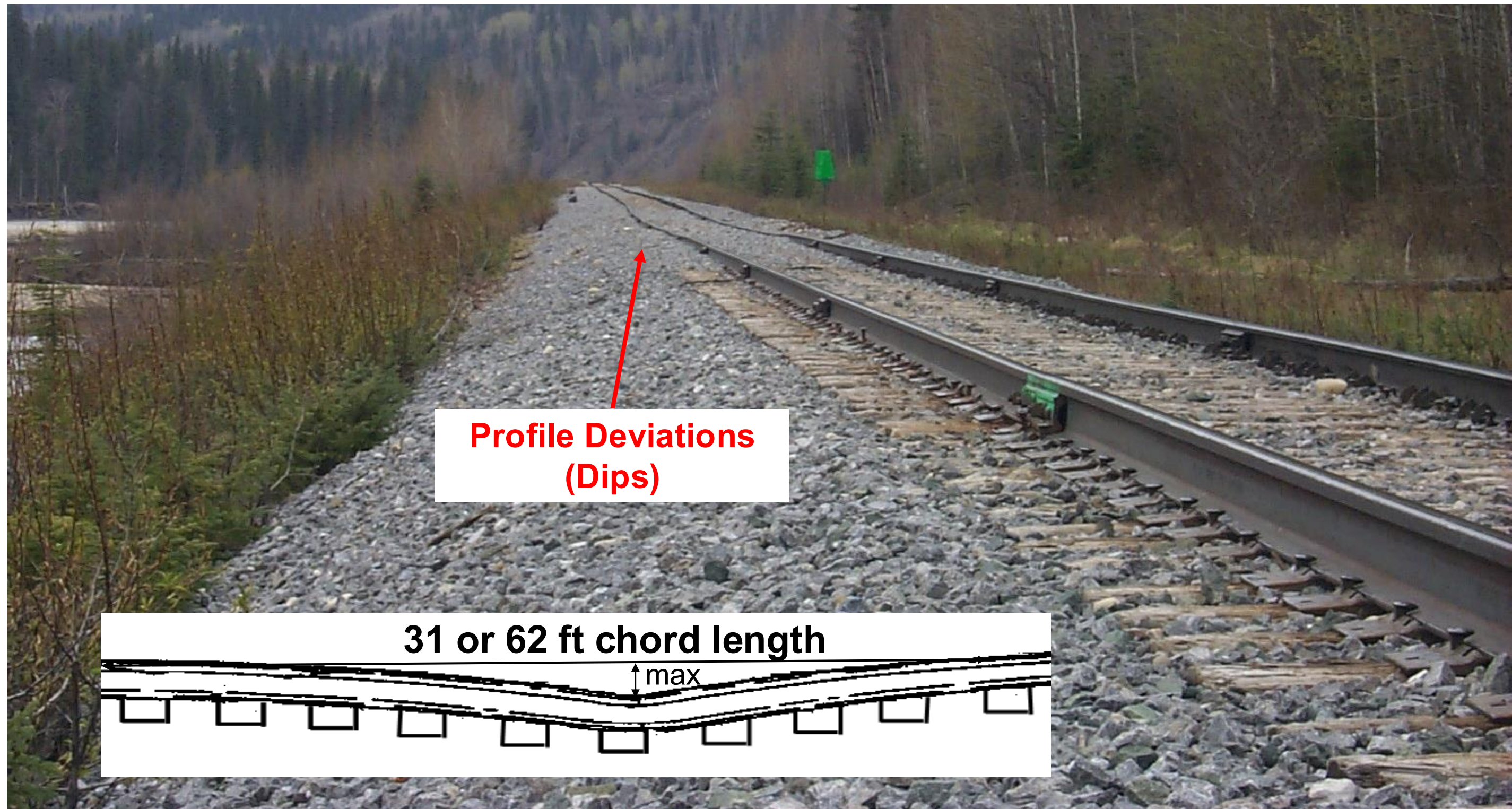
§ 213.55 Track alinement

55(a) Except as provided in paragraph (b) of this section, alinement may not deviate from uniformity more than the amount prescribed in the following table:

Class of Track	Tangent Track	Curved Track	
	The deviation of the mid-offset from a 62-foot line ¹ may not be more than—(inches)	The deviation of the mid-ordinate from a 31-foot chord ² may not be more than—(inches)	The deviation of the mid-ordinate from a 62-foot chord ² may not be more than—(inches)
1	5	N/A ³	5
2	3	N/A ³	3
3	1¾	1¼	1¾
4	1½	1	1½
5	¾	½	⅝



Track Surface (Profile) Deviation



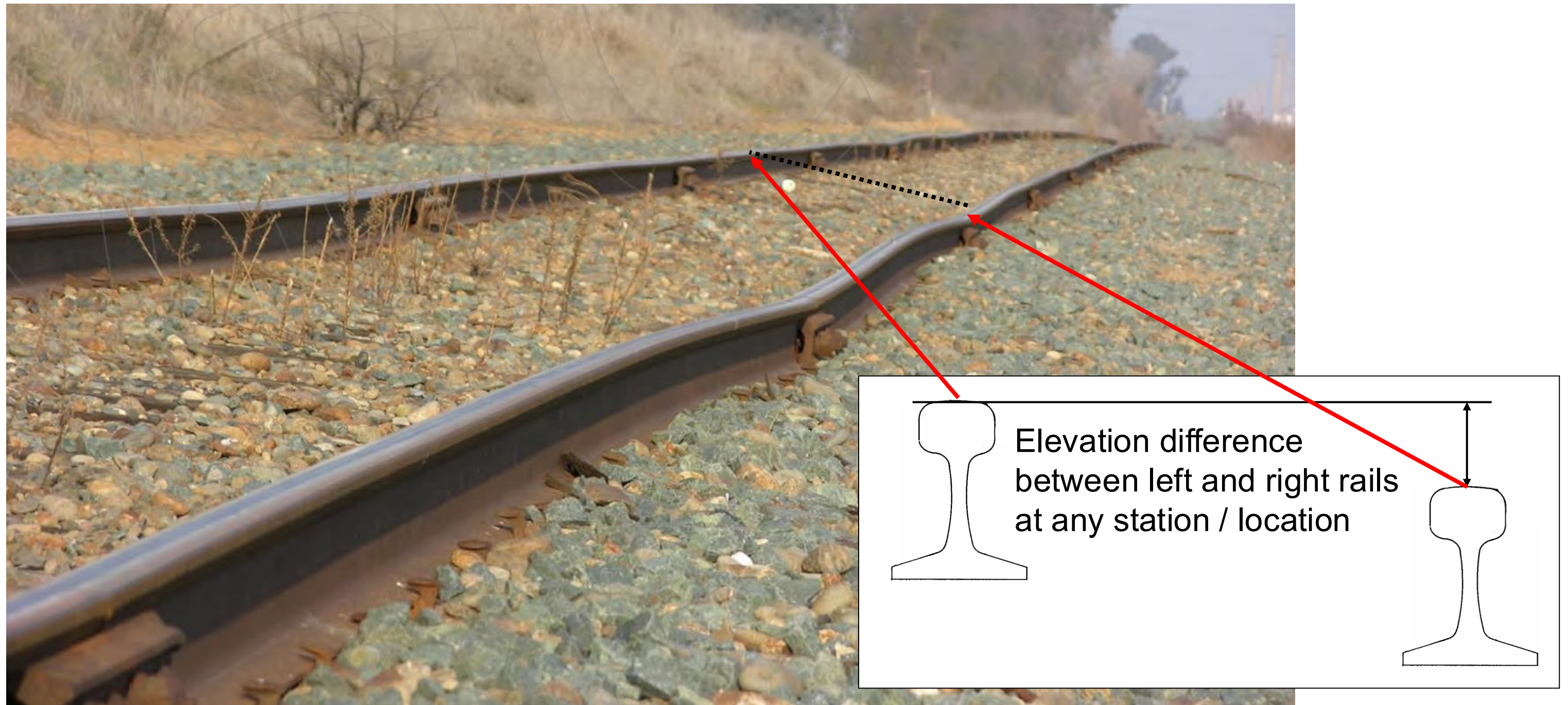
Track Surface (Profile)

§ 213.63 Track Surface

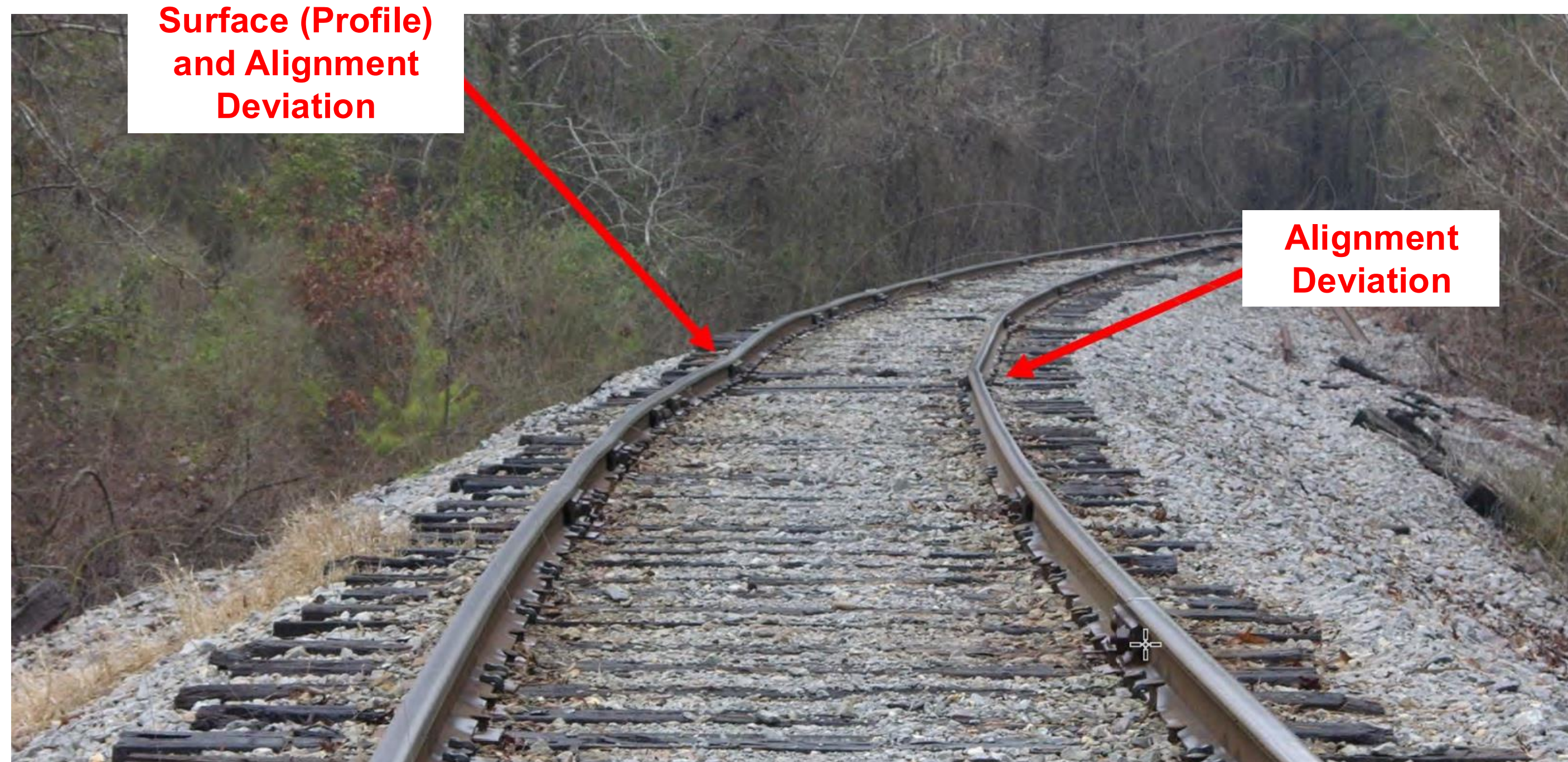
63(a) Except as provided in paragraph (b) of this section, each track owner shall maintain the surface of its track within the limits prescribed in the following table:

Track surface (inches)	Class of track				
	1	2	3	4	5
The runoff in any 31 feet of rail at the end of a raise may not be more than	3½	3	2	1½	1
The deviation from uniform profile on either rail at the mid-ordinate of a 62-foot chord may not be more than ...	3	2¾	2¼	2	1¼
The deviation from zero crosslevel at any point on tangent or reverse crosslevel elevation on curves may not be more than	3	2	1¾	1¼	1
The difference in crosslevel between any two points less than 62 feet apart may not be more than ^{*1, 2}	3	2¼	2	1¾	1½
*Where determined by engineering decision prior to June 22, 1998, due to physical restrictions on spiral length and operating practices and experience, the variation in crosslevel on spirals per 31 feet may not be more than.....	2	1¾	1¼	1	¾

Crosslevel Deviation



Combined Surface & Alignment Deviations



FRA Inspection Requirements

§ 213.233 Track inspections

233(c) Each track inspection shall be made in accordance with the following schedule:

Class of Track	Type of Track	Required Frequency
<i>Excepted track and Class 1, 2, and 3 track</i>	<i>Main track and sidings</i>	<i>Weekly with at least 3 calendar days interval between inspections, or before use, if the track is used less than once a week, or twice weekly with at least 1 calendar day interval between inspections, if the track carries passenger trains or more than 10 million gross tons of traffic during the preceding calendar year.</i>
<i>Excepted track and Class 1, 2, and 3 track</i>	<i>Other than main track and sidings</i>	<i>Monthly with at least 20 calendar days interval between inspections.</i>
<i>Class 4 and 5 track</i>	<i>.....</i>	<i>Twice weekly with at least 1 calendar day interval between inspections.</i>

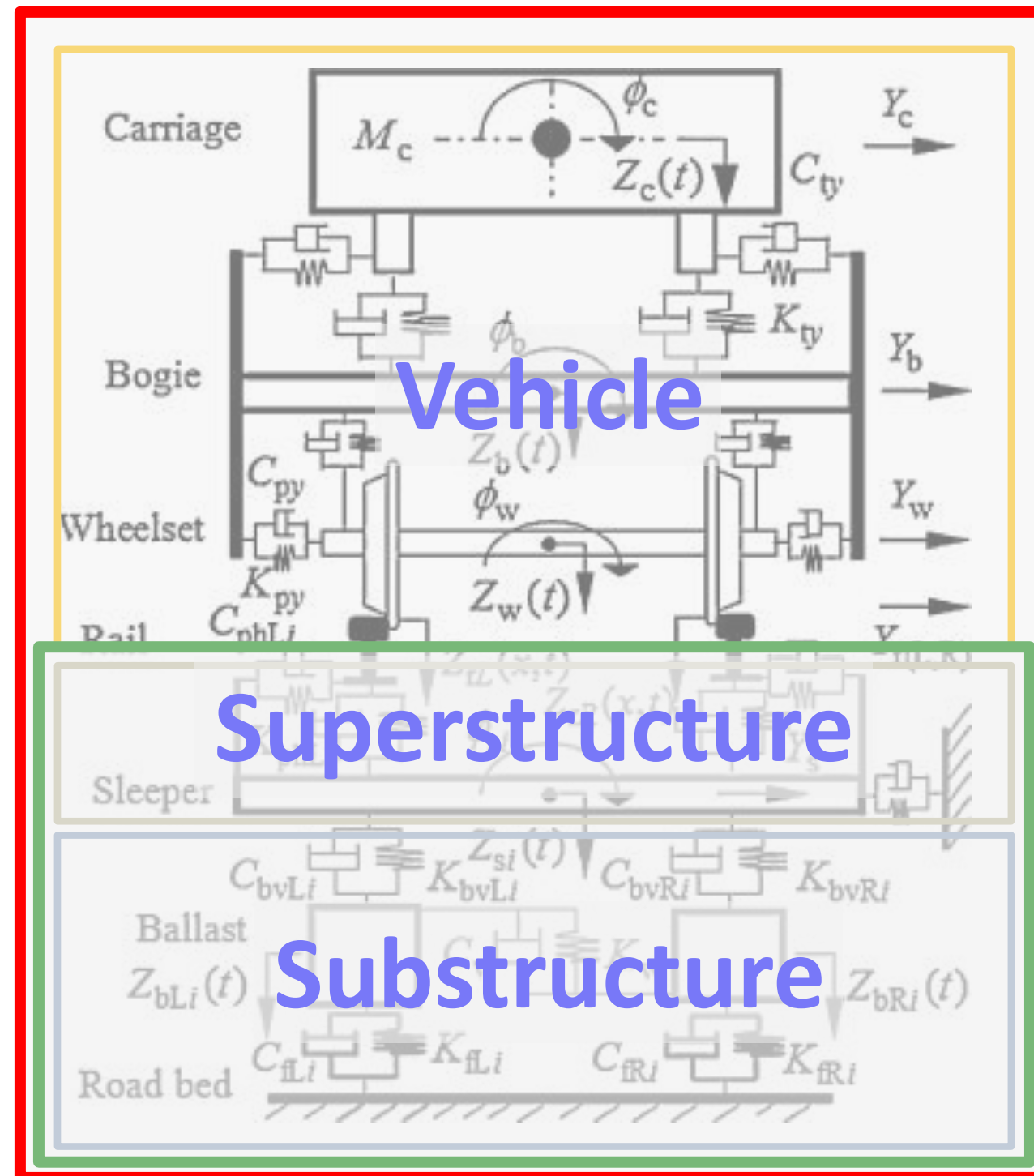
Other FRA Inspection Requirements

- Automated inspection of track constructed with concrete crossties (CFR 213.234)
- Inspection of switches, track crossings and lift rail assemblies or other transition devices on movable bridges (CFR 213.235)
- Inspection of rail (CFR 213.237 and 213.339)
- Special inspections (CFR 213.239)
- Automated vehicle-based inspection systems (CFR 213.333)

See the FRA ***“Track and Rail and Infrastructure Integrity Compliance Manual – Volume II”*** for more details

The Track & Vehicle as a System

Understanding System Dynamics

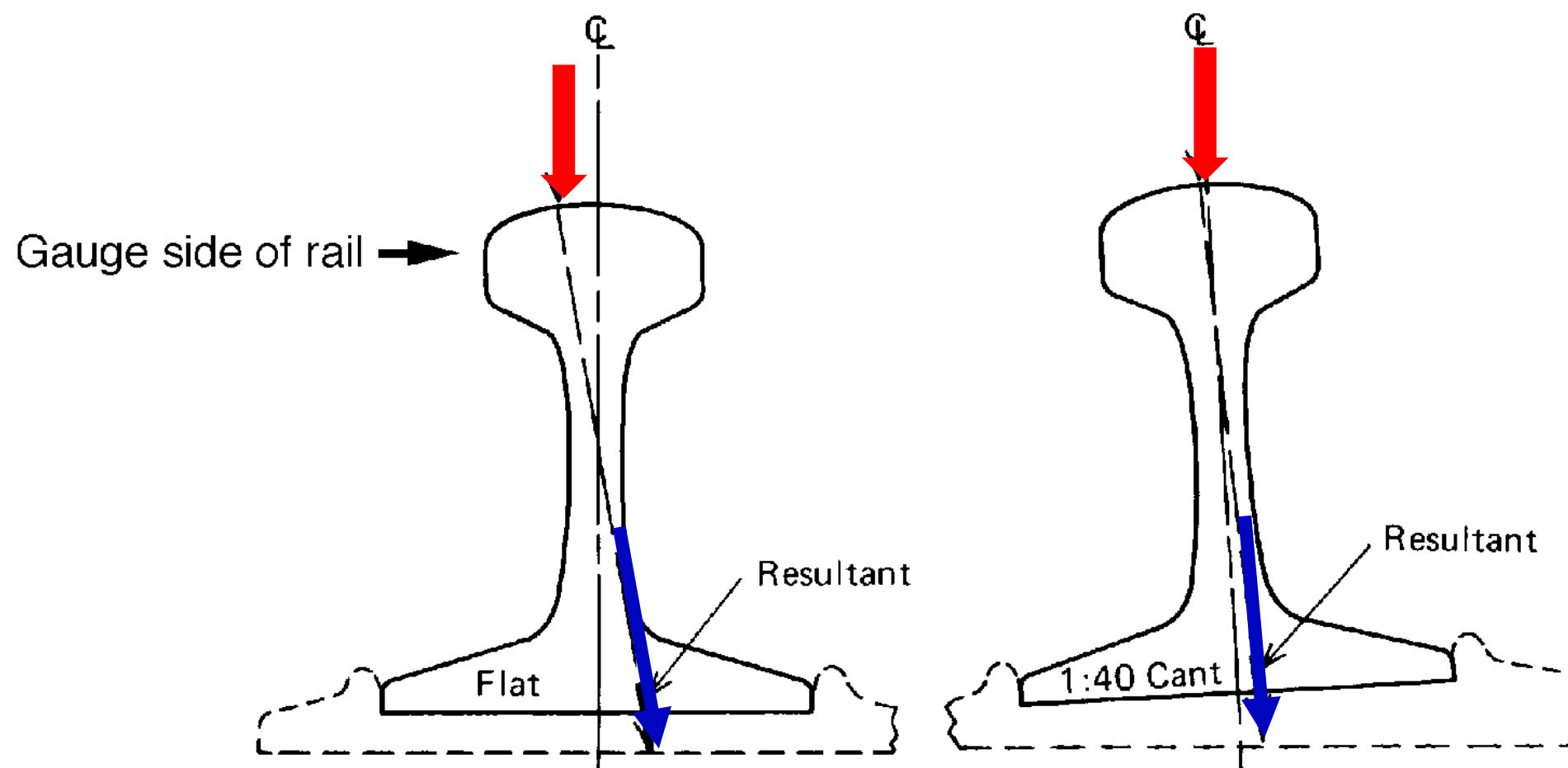


Wei Li, Guangwen Xiao, Zefeng Wen, Xinbiao Xiao, Xuesong Jin
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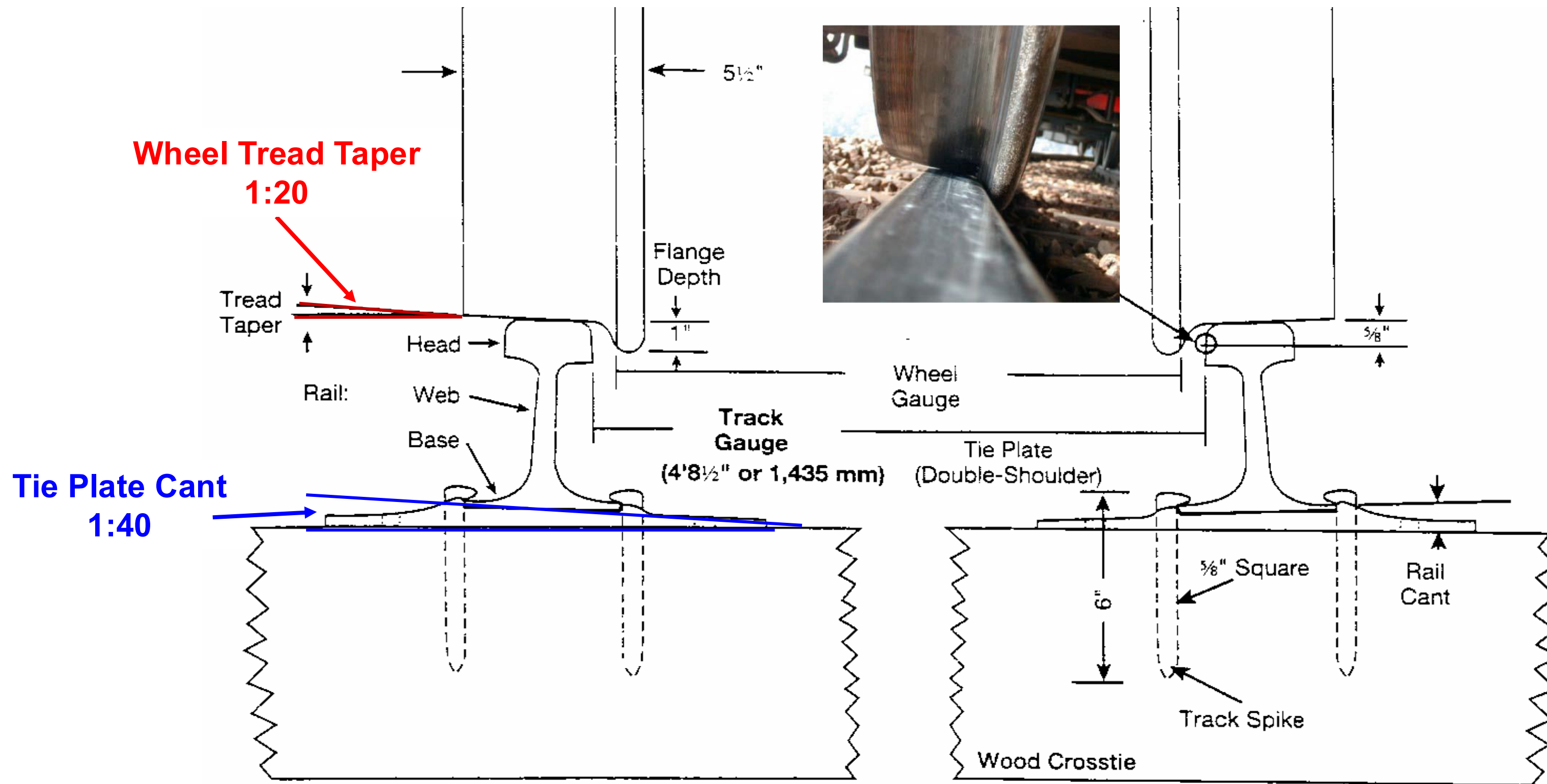
Rail Cant

(Rail Seat Inclination)

- Because the contact patch with a normal, tapered wheel is not at the center of the head of the rail, the rail is canted inward (toward the gauge side of the track) to give a better, more stable resultant load path through the web to the base.
- Typical North American practice is from 1:20 to **1:40 cant**



Wheel Tread Taper and Rail Cant

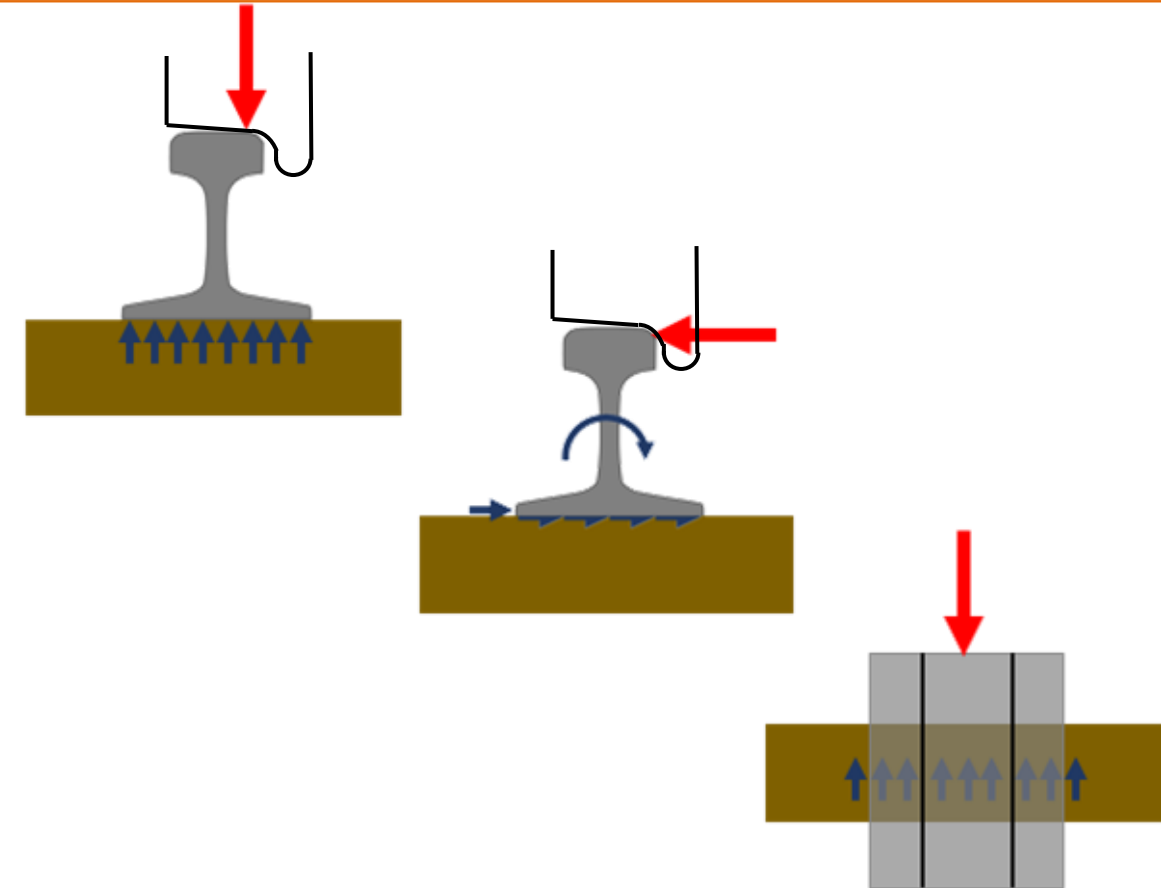


The Track Loading Environment Basics

Forces Acting on the Track

- Load Direction and Orientation:

- **Vertical** (P, Q or V)
- **Lateral** (Y or L)
- **Longitudinal** (T)



- Primary Types of Loading:

- **Static** → Weight of vehicle at rest (level track)
- **Dynamic** → Wheel rotation, minor track and wheel irregularities
 - Largely a function of the railcar in motion
- **Impact** → Often related to discontinuities
 - Vehicle wheel-induced or track-related

Common North American Freight Car Sizes and Loads

Nominal Capacity (cargo weight)	Gross Rail Load (GRL) (vehicle PLUS cargo weight)	Axle Load (4 axles)	Wheel Load (8 wheels)
140,000 lbs (70 ton)	220,000 lbs	55,000 lbs (27.5 tons)	27,500 lbs
200,000 lbs (100 ton)	263,000 lbs	65,750 lbs (32.9 tons)	32,875 lbs
220,000 lbs (110 ton)	286,000 lbs	71,500 lbs (35.8 tons)	35,750 lbs
250,000 lbs (125 ton)	315,000 lbs	78,750 lbs (39.4 tons)	39,375 lbs

Common Design Load

“Heavy Axle Load” (HAL)

Vertical Forces Acting on the Track

- Vertical (P , Q or V)

$$P_{total} = (P_{static} + P_{centering} + P_{wind}) + P_{dynamic}$$

quasi-static forces

where:

P_{static} = static wheel load measured on tangent track

$P_{centering}$ = wheel load due curvature and cant

P_{wind} = increase in wheel load due to wind

$P_{dynamic}$ = dynamic wheel load components from:

sprung mass (0-20 Hz); unsprung mass (20-125 Hz); & corrugations,
weld, wheel flats (0-2000 Hz)

Impact Loads

- Caused by wheel defects (flat spots), rail discontinuities, track transitions, etc.
 - Generally higher magnitude than cyclic dynamic loads
 - Flat or out-of-round wheels cause impact loadings in excess of 3X Static at the railhead and 4.5X Static in the subgrade
- High frequency vibrations referred to as P1 and P2 forces
 - P1 Force: High impact, short duration (rail, concrete ties)
 - P2 Force: High impact, long duration (ballast, track geometry)
 - Note: P1 forces can be up to about 40% higher than P2 forces

Lateral Forces Acting on the Track

- Lateral (Y or L)

$$Y_{total} = (Y_{flange} + Y_{centering} + Y_{wind}) + Y_{dynamic}$$

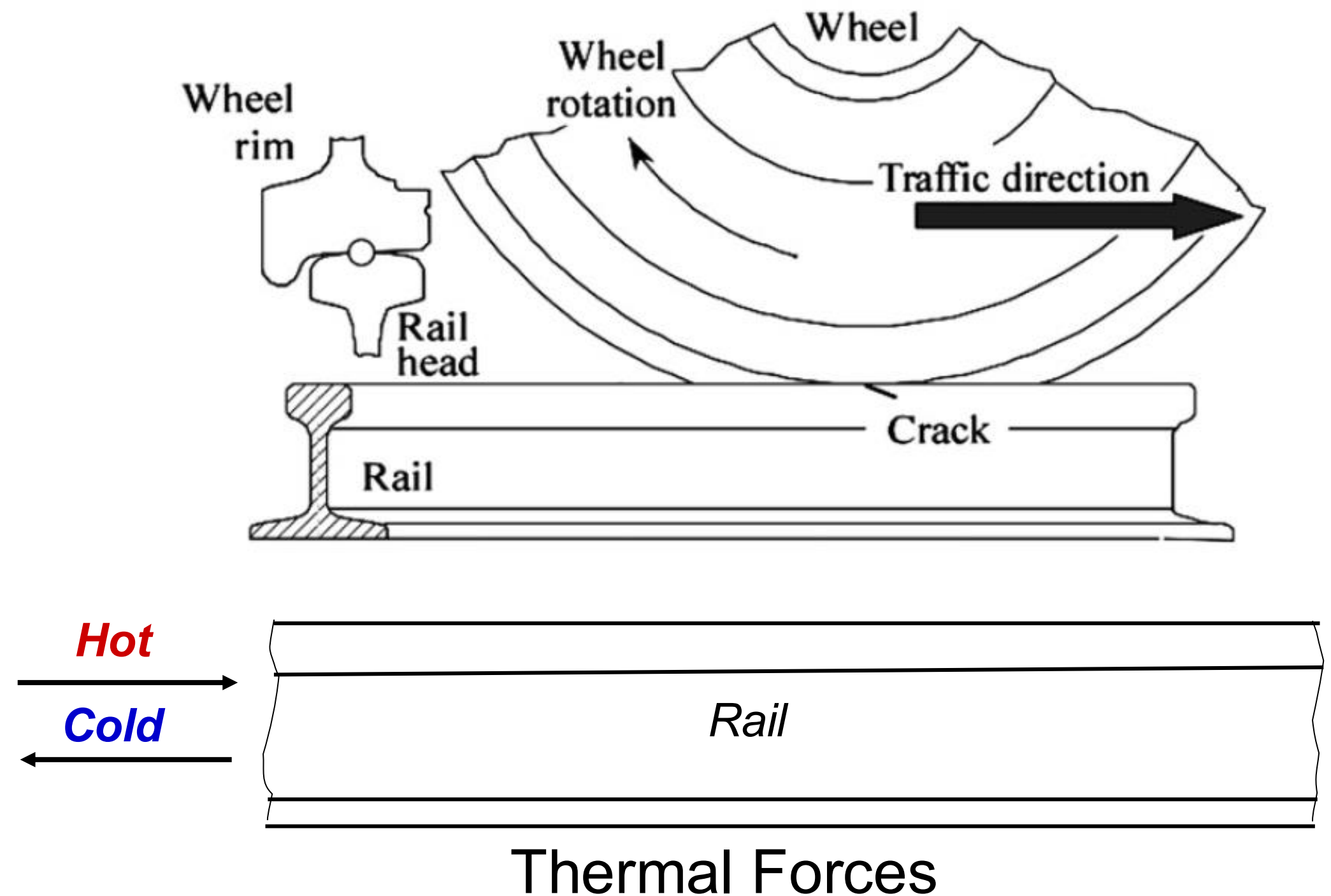
quasi-static forces

where:

- Y_{flange} = lateral load from flanging on outer rail in curve
- $Y_{centering}$ = lateral force due to non-compensated centrifugal force
- Y_{wind} = lateral force due to cross wind
- $Y_{dynamic}$ = dynamic lateral force component due to hunting or track alignment deviations

Longitudinal Force Acting on the Track

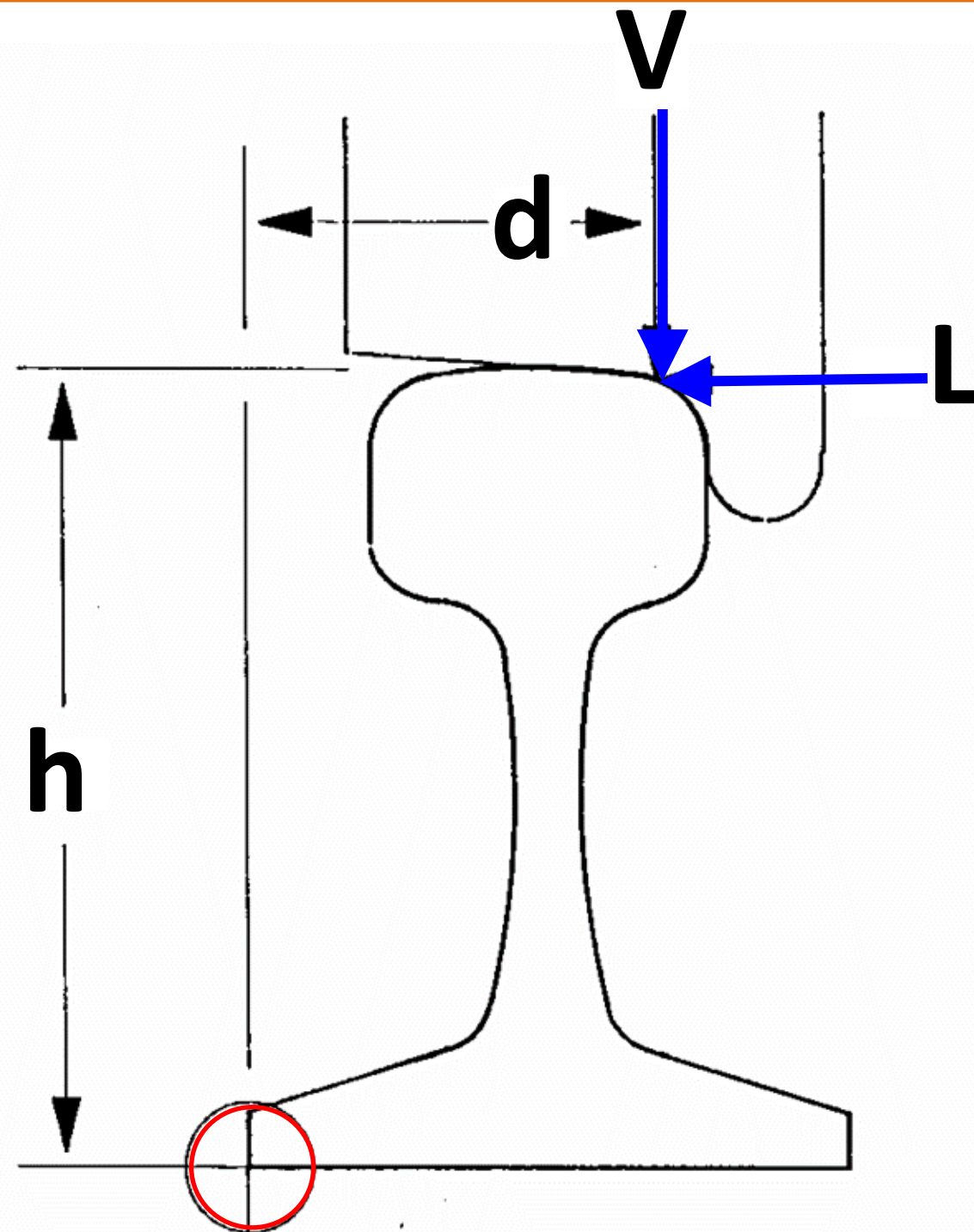
- Thermal Force
- Traction Force
- Braking Force



Track Forces and their Influence

Force Direction	Load Magnitude (Single Wheel)	Load Amplification	Load Sharing (Stiffness)
Vertical	32,000 – 40,000 lbs	1.0 - 1.5x Dynamic 2.0 - 4.0x Impact	5-7 Crossties
Lateral	Very low up to 10,000 – 20,000 lbs	1.0 - 2.0x	3-4 Crossties
Longitudinal	Very Low to 20,000 lbs	Negligible	60-300 Crossties (100-500 ft)

L/V Ratio and Derailment Hazard



Rail can roll when: $\frac{L}{V} > \frac{d}{h}$

Ratio	Degree of Hazard
0.68	The resultant (making an angle of approximately $33^{\circ}20'$ with the vertical load) passes outside the base of the rail, indicating initial instability; an unrestrained rail may overturn.
0.75	A worn wheel flange may climb a worn rail.
0.82	The flange disengages from the rail; an outside wheel may lift from the rail on curves.
1.29	Derailing condition; wheel will climb a new rail; a new wheel lowers the wheel climb threshold.

Back to our case study site...

What do you see?



- Rail
- Crossties and fasteners
 - Timber ties, cut spikes, and rail anchors (EOTA)
- Ballast
- Horizontal curvature
 - Superelevation
- Proximity to grade crossing or other fixed objects?
- Any other items of interest?



CONTACT

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